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Co-optimizing of grouping and aggregated bidding for multiple wind farms under uncertainties

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HIGHLIGHTS

- Embeds complementarity metrics (over/under outcomes and tail dependence) to form groups.
- Co-optimizes wind-farm grouping and aggregated bidding under price and wind uncertainty.
- Tail-consistent, budget-balanced settlement allocates energy and risk-savings fairly.

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ABSTRACT

Market-based day-ahead bidding is increasingly used to support renewable integration, but wind producers remain exposed to imbalance costs and revenue volatility due to forecast errors. While many risk-mitigation approaches rely on external flexibility resources, systematic methods for internal coordination among multiple wind farms that jointly address grouping, bidding, and benefit allocation remain limited. This paper proposes an uncertainty-driven framework for cooperative wind-farm grouping and aggregated day-ahead bidding under exogenous day-ahead prices and wind-power uncertainty. Complementarity among wind farms is quantified and verified using scenario-based over- and under-generation patterns together with tail dependence, while the grouping and bidding decisions favor complementary combinations through a risk-aware objective on aggregated imbalance losses. Beyond the grouping and bidding co-optimization layer, the main contribution lies in a tail-consistent and budget-balanced settlement mechanism that translates the uncertainty-reduction benefits created by cooperative aggregation into implementable member-level payments. Energy revenues are allocated through a stable capacity-proportional rule, and the risk-savings pool is redistributed using scenario-level imbalance attribution and tail-responsibility weights. Numerical experiments based on exogenous day-ahead price profiles and simulated wind data show that the proposed framework reduces imbalance risk, improves risk-adjusted revenues, and outperforms benchmark allocation rules. The case study also indicates that the resulting settlement remains transparent and practically attractive at the member level, while sensitivity analyses confirm robustness to key risk parameters.

1. Introduction

The global trend toward reducing carbon dioxide emissions is accelerating the growth of various renewable energy sources in power systems, with wind power accounting for an increasing share of new generation capacity [4,5,29]. To facilitate large-scale integration of renewable energy, electricity markets have been restructured to improve the valuation of flexibility and to support renewable generation through transparent pricing and complementary market mechanisms. In parallel, time-varying market price signals increasingly affect the revenues of renewable generators and tighten the link between operational uncertainty and bidding outcomes. Within this context, wind power has

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Nomenclature

Sets and indices

$\mathcal{I}$	Set of wind farms, index i.
$\mathcal{G}$	Set of groups, index g.
$\mathcal{H}$	Set of operating hours, index h.
$\mathcal{S}$	Set of scenarios, index s.
$\mathcal{N}$	Set of wind farms in the cooperative game.
$\mathcal{E}$	Set of incompatible wind-farm pairs, where $(i,j) \in \mathcal{E}$ means wind farms i and j are not allowed to be assigned to the same group.
$T_{g,h}$	Tail set of scenarios for group g at hour h under RU–CVaR.

Variables

$z_{i,g}$	Binary variable, equal to 1 if wind farm i is assigned to group g.
$Q_{g,h}$	Day-ahead bid quantity of group g at hour h.
$\theta_{i,g,h}$	Share of group bid $Q_{g,h}$ allocated to wind farm i.
$\xi^{\downarrow}_{g,h,s}$	Deviation variable for shortfall of group g at hour h in scenario s.
$\xi^{\uparrow}_{g,h,s}$	Deviation variable for surplus of group g at hour h in scenario s.
$q_{i,h}$	Individual day-ahead bid of wind farm i at hour h (benchmark case).
$\tau_{g,h}$	RU–CVaR auxiliary variable at group g and hour h.
$u_{g,h,s}$	RU–CVaR slack variable associated with loss $L_{g,h,s}$.
$R^{\text{energy}}_{g,h}$	Energy revenue of group g at hour h.
$r^{\text{energy}}_{i,h}$	Energy revenue allocated to wind farm i at hour h.
$r^{\text{energy,day}}_{i}$	Daily energy revenue allocated to wind farm i.
$RS_{g,h}$	Risk-savings pool of group g at hour h.
$r^{\text{risk}}_{i,h}$	Risk-savings payment to wind farm i at hour h.
R^{tot}_{i}	Total daily settlement of wind farm i.
$\Delta U_{g,h,s}$	Shortfall (under-production) of group g at hour h in scenario s.
$\Delta O_{g,h,s}$	Surplus (over-production) of group g at hour h in scenario s.
$L_{g,h,s}$	Imbalance loss of group g at hour h in scenario s.
$L^{\text{ind}}_{i,h,s}$	Imbalance loss of wind farm i under individual bidding in scenario s.
$B^{\text{share}}_{i,h}$	Capacity-based baseline share of group bid for wind farm i at hour h.
$a_{i,h,s}$	Under-production indicator of wind farm i relative to $B^{\text{share}}_{i,h}$ in scenario s.
$b_{i,h,s}$	Over-production indicator of wind farm i relative to $B^{\text{share}}_{i,h}$ in scenario s.
$\delta U_{i,h,s}$	Share of group shortfall attributed to wind farm i in scenario s.
$\delta O_{i,h,s}$	Share of group surplus attributed to wind farm i in scenario s.
$L^{\text{contrib}}_{i,h,s}$	Loss contribution of wind farm i at hour h in scenario s.
$\omega_{i,g,h}$	Tail-responsibility weight of wind farm i in group g at hour h.
$v_h(S)$	Coalition worth of coalition $S \subseteq \mathcal{N}$ at hour h (risk-savings).
$\phi^{\text{Sh}}_{i,h}$	Shapley value of wind farm i at hour h.
$c_{i,h}$	Normalized tail-loss contribution of wind farm i at hour h.
$u_{i,h}$	Per-MW risk-savings payout of wind farm i at hour h.
TAC_h	Tail-alignment coefficient at hour h.
IRmin_h	Minimum per-MW risk margin among members at hour h.
EFmax_h	Maximum envy-freeness gap at hour h.
NEF_h	Normalized envy-freeness ratio at hour h.
Gini_h	Gini coefficient of per-MW payouts at hour h.

Parameters

x_i, y_i	Coordinates of wind farm i.
$D_{i,j}$	Distance between wind farms i and j.
ρ_0	Base correlation level between wind farms.
ℓ	Correlation length in the exponential distance-decay kernel.
$R_{i,j}$	Entry of the spatial correlation matrix between wind farms i and j.
k_i	Weibull shape parameter of wind speed at wind farm i.
$\lambda_{i,h}$	Weibull scale parameter of wind speed at wind farm i and hour h.
$\lambda^{\text{base}}_{i}$	Baseline scale parameter of wind speed at wind farm i.
λ^{amp}_{i}	Amplitude of diurnal modulation of the scale parameter at wind farm i.
C_i	Installed capacity of wind farm i.
$\gamma_{i,h}$	Johnson SU location parameter of forecast residuals at wind farm i, hour h.
$\delta_{i,h}$	Johnson SU shape parameter of forecast residuals at wind farm i, hour h.
$\xi_{i,h}$	Johnson SU shift parameter of forecast residuals at wind farm i, hour h.
$\lambda^{\text{JSU}}_{i,h}$	Johnson SU scale parameter of forecast residuals at wind farm i, hour h.
$\pi_{i,h}$	Exogenous day-ahead price of wind farm i at hour h.
$\bar{\pi}_{g,h}$	Capacity-weighted group day-ahead price at hour h.
$\kappa^{\downarrow}_{h}$	Penalty coefficient for shortfall (under- generation) at hour h.
$\kappa^{\uparrow}_{h}$	Penalty coefficient for surplus (over- generation) at hour h.
λ	Risk-aversion weight in the objective.
α	Confidence level of CVaR.
w_s	Probability weight of scenario s.
S	Number of scenarios used in the SAA estimator.
B, c	Constants in the finite-sample bound for the CVaR estimator.
v_α	Value-at-Risk (VaR) at confidence level α.
$\psi_{i,h}$	Generic allocation weight for distributing $RS_{g,h}$ at hour h.
$\psi^{\text{tail}}_{i,h}$	Allocation weight under the tail-consistent rule.
$\psi^{\text{allscen}}_{i,h}$	Allocation weight under the all-scenarios rule.
$\psi^{\text{cap}}_{i,h}$	Allocation weight under the capacity-based rule.
$\psi^{\text{eng}}_{i,h}$	Allocation weight under the energy-based rule.
$\psi^{\text{eq}}_{i,h}$	Allocation weight under the equal-share rule.
$\psi^{\text{shapley}}_{i,h}$	Allocation weight based on the Shapley value.
$E_{i,h}$	Expected energy production of wind farm i at hour h.

Random variables and scenario-dependent quantities

$p_{i,h,s}$	Real power output of wind farm i at hour h in scenario s.
$P_{g,h,s}$	Real power output of group g at hour h in scenario s.
$\hat{p}_{i,h}$	Day-ahead power forecast of wind farm i at hour h.
$r_{i,h}$	Forecast residual of wind farm i at hour h.
$r_{i,h,s}$	Sampled residual of wind farm i at hour h in scenario s.

received sustained attention due to its variability, forecast errors, and spatial heterogeneity, all of which complicate bidding and revenue adequacy while creating new opportunities for portfolio design and risk management [24].

The uncertainty of renewable generation has prompted wind power producers to continuously improve their bidding strategies. Stochastic programming offers an intuitive and systematic framework to address such uncertainty, where scenario-based bi-level formulations coordinate day-ahead and real-time decisions while incorporating CVaR terms to control downside risk [8]. Building on this foundation, pay-as-bid mechanisms within the same bi-level structure enable producers to internalize their price impact while maintaining exposure control [1]. To enhance temporal representativeness, multi-stage intra-day models link consecutive trading sessions, allowing participants to exploit arbitrage opportunities and improve inter-temporal flexibility under volatile conditions [6]. Further extensions integrate reliability requirements directly into the stochastic framework through chance-constrained formulations, which preserve tractability under suitable distributional assumptions and capture the coordinated participation of multiple wind farms in both energy and reserve markets [12]. Recent studies also show that correlated wind production can be incorporated directly into offering strategies, for example in wind-storage systems where dependence-aware stochastic modeling is used to improve bidding robustness under coupled renewable uncertainty [10].

Nevertheless, the performance of stochastic models often depends on the accuracy of scenario generation and assumed probability distributions, motivating the development of robust counterparts. To address this issue, robust optimization hedges against forecast uncertainty through uncertainty sets, while adaptive recourse strategies enhance operational flexibility and jointly schedule complementary assets such as storage [2]. However, its high computational cost and excessive conservatism often constrain practical deployment in large-scale multi-period bidding. To combine the strengths of stochastic and robust frameworks, hybrid stochastic–robust formulations balance expected-profit maximization with worst-case resilience while considering practical factors such as degradation costs [25]. More recently, distributionally robust optimization bridges these approaches by optimizing over a series of reasonable probability distributions, where segmented ambiguity sets capture heterogeneous forecast errors and enable a more practical compromise between computational efficiency and robustness [20].

Another direction of research reduces bidding risk by aggregating wind power with other flexible resources. Typical examples include joint operation with energy storage systems (ESS) [31], pumped-hydro units [15], power-to-gas facilities [34], and electric-vehicle fleets [17,33], which can enhance portfolio flexibility and reduce forecasting risk. Under the virtual power plant (VPP) concept, wind power is integrated with other dispatchable assets [7,16,23], and their energy and reserve bids are optimized together to reduce power imbalances and imbalance penalties. A recent representative example is the shared multi-energy storage framework in [3], where wind producers rent battery and hydrogen storage capacity and jointly participate in the day-ahead, balancing, and hydrogen markets under a CVaR-based bidding model. Compared with such storage-assisted approaches, the present work focuses on a different mechanism, namely reducing imbalance risk through internal coordination among multiple wind farms themselves, without relying on external storage assets or additional market layers. Related work has also examined transaction benefit allocation in shared-storage settings, for example through two-stage robust transaction and benefit-allocation models for new energy power stations with shared energy storage [13]. However, such allocation is still built on external storage-supported coordination rather than on endogenous grouping and internal uncertainty compensation among wind farms themselves. Some studies also couple energy markets with environmental products, such as renewable certificates [9], green electricity labels [28], and carbon policies [30]. Therefore, wind producers can obtain extra income while meeting policy targets. In most of these works, however, wind power uncertainty is absorbed by adding external flexible resources or extra market layers, rather than by coordinating wind farms themselves to offset uncertainty through internal cooperation.

Despite extensive research on wind participation in electricity markets, most studies still handle uncertainty by coupling wind farms with external flexible resources such as energy storage or power-to-gas units. These schemes can reduce imbalances, but they require large capital investment, extra operation and maintenance, and complex coordination among different owners. From the system point of view, wind power is often seen as difficult energy because its variability has to be absorbed by storage, flexible generators, or demand response, which could otherwise serve other purposes. If part of this variability could be absorbed inside a group of wind farms themselves, the burden on external resources would be reduced and what is usually treated as waste variability could become a useful hedging effect. In particular, when heterogeneous wind farms are jointly scheduled and settled, their forecast errors can partially cancel out, leading to smaller aggregated deviations and reduced exposure to extreme imbalance losses. However, systematic models for such internal coordination are still limited. Existing works seldom optimize wind-farm grouping based on cross-farm dependence and practical participation constraints, nor do they link this grouping to joint bidding and settlement rules that can be implemented in real markets.

Table 1 compares representative related studies with the proposed framework across uncertainty treatment, key decision variables, target markets, and optimization framework. Existing studies mainly mitigate wind uncertainty by coupling wind power with external flexible resources such as shared storage, pumped storage, power-to-gas, or broader virtual power plant portfolios. In most of these works, the asset portfolio is fixed a priori, and the main decision is bidding and dispatch. By contrast, the present study addresses a different question. Multiple wind farms are coordinated directly, the group membership itself is optimized, and the value created by uncertainty reduction is redistributed through a tail-consistent settlement rule. Therefore, the main gap addressed here is the lack of a unified framework that jointly determines endogenous wind-farm grouping, aggregated day-ahead bidding, and member-level allocation of uncertainty-reduction benefits.

Accordingly, this paper focuses on three linked research questions. The first is which wind farms should cooperate under uncertainty. The second is how each resulting group should determine its aggregated day-ahead bid. The third is how the uncertainty-reduction benefits created by cooperation should be allocated among members so that participation remains attractive and internally consistent. This paper develops a unified framework that forms groups based on cross-farm dependence and participation constraints, determines their aggregated bids in the day-ahead market, and settles the resulting energy revenues and risk-saving benefits inside each group. A key methodological novelty lies in the settlement layer, which makes the economic value of uncertainty reduction through cooperative aggregation implementable at the member level. The main contributions of this paper are as follows:

1. A co-optimization framework is proposed to jointly determine wind-farm grouping and aggregated day-ahead bidding for multiple wind farms under exogenous day-ahead price and wind-speed uncertainty. The framework addresses coordinated bidding among heterogeneous wind farms and remains tractable under realistic operational constraints. Complementary combinations are favored endogenously because aggregated deviation risks and extreme imbalance losses are penalized at the group level.
2. A quantitative method is developed to identify each wind farm's role in uncertainty reduction when it is grouped with others. Scenario-based over- and under-generation outcomes together with tail dependence are used to quantify and verify complementarity between wind farms. The co-optimization captures such complementarity endogenously through its risk-aware objective on aggregated imbalance losses, rather than through explicit complementarity constraints.
3. A tail-consistent and budget-balanced settlement mechanism is developed as the core contribution of this work to allocate the economic benefits created by uncertainty reduction in a fair and transparent manner. Energy revenues are allocated through a stable capacity-proportional rule, while the group-level risk-savings pool is redistributed according to scenario-level imbalance attribution and tail-responsibility weights that are consistent with the risk objective. In this way, uncertainty-reduction benefits

Table 1
Comparison of representative recent studies and the proposed framework.

Reference	Coordination mode	Uncertainty treatment	Key decision variables	Target market(s)	Optimization framework	Endogenous grouping	Explicit benefit allocation
[23]	Renewable-based VPP with fixed assets	Renewable uncertainty with risk-aware bidding	Bids, dispatchable-load scheduling	Day-ahead	Optimization-based bidding model	No	No
[31]	Wind power with shared energy storage	Wind uncertainty in day-ahead and real-time operation	Bids, real-time scheduling, storage charge/discharge	Day-ahead and real-time	Two-stage scheduling / optimization	No	No
[10]	Wind-storage system with fixed portfolio	Correlated wind production	Offering strategy, storage scheduling	Day-ahead	Stochastic optimization	No	No
[33]	Wind power coordinated with large-scale EVs	Wind and market uncertainty	Energy bids, regulation participation, EV power allocation	Day-ahead and frequency regulation	Coordinated optimization	No	No
[15]	Wind power and pumped storage coordination	Multiple uncertainties	Joint declaration and coordinated dispatch	Power market	Risk-constrained optimization	No	No
[7]	Virtual power plant with heterogeneous resources	Real-time operational uncertainty	Real-time dispatch and power disaggregation	Real-time	Optimization-based operation strategy	No	No
[13]	New energy stations with shared energy storage	Two-stage robust uncertainty treatment	Transaction decisions, storage usage, benefit allocation	Electricity-related transactions	Two-stage robust optimization	No	Yes
[3]	Wind plants with rented shared multi-energy storage	CVaR-based risk management under wind uncertainty	Day-ahead bids, balancing decisions, hydrogen participation, storage usage	Day-ahead, balancing, and hydrogen	MILP with CVaR	No	No
[16]	Internal and external coordinated VPP portfolio	Distributionally robust renewable uncertainty	External bids, internal dispatch decisions	Day-ahead spot and ancillary services	Two-layer distributionally robust optimization	No	No
This paper	**Direct internal coordination among multiple wind farms**	**Scenario-based wind uncertainty with CVaR-based tail-risk penalization**	**Group membership, aggregated day-ahead bids, and member-level settlement**	**Day-ahead**	**MILP co-optimization with ex-post tail-consistent settlement**	**Yes**	**Yes**

created at the group level are translated into implementable member-level payments. The settlement rule is further supported by internal consistency, budget balance, and a conditional local incentive-compatibility property, while participation attractiveness (individual rationality) is evaluated in the case study under practical market settings.

The rest of the paper is organized as follows. Section 2 presents the process of the data generation and uncertainty fitting. In Section 3, the co-optimization model is proposed to determine the group formation and aggregated bids in the day-ahead market. Section 4 presents the tail-consistent settlement rule. Section 5 conducts a numerical study to verify the effectiveness of the mechanism. Section 6 concludes the paper.

2. Wind power data generation and uncertainty modeling

Wind power exhibits significant uncertainty and variability, which directly affects imbalance settlements and tail losses in electricity markets. In this section, a scenario-based data generation procedure is adopted to provide consistent uncertainty inputs for the proposed optimization and settlement analysis. Wind speeds are first sampled with diurnal variation and cross-farm spatial dependence so that realistic co-movements across sites are preserved. The sampled wind speeds are then mapped to power outputs and corresponding forecasts, from which forecast residuals are obtained to represent the uncertainty relevant to market participation. Finally, residual scenarios with non-Gaussian and heavy-tailed characteristics are generated and coupled across wind farms to form a spatially consistent scenario set. These scenarios are used in the subsequent sections to evaluate grouping decisions, aggregated bids, and tail-risk outcomes. To keep this section concise, the explicit Johnson SU form, parameter interpretation, the Gaussian-copula residual-scenario generation steps, and the baseline assumption of uniform scenario weights are provided in Appendix A.

For each wind farm i, a time series of hourly wind speeds is sampled with diurnal variation, and cross-farm dependence is imposed through a distance-decay correlation structure. Specifically, the Euclidean distance between wind farms i and j is

$$D_{i,j} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}, \tag{1}$$

and the corresponding spatial correlation matrix is defined by

$$R_{i,j} = \rho_0 \exp\left(-D_{i,j}/\ell\right), \tag{2}$$

where $\rho_0 \in [0,1]$ and $\ell > 0$ are the base correlation and correlation length. A Gaussian-copula sampling framework is then used to generate correlated hourly wind-speed samples that preserve $\boldsymbol{R}$ while maintaining farm-wise marginals [22]. Diurnal variation is represented by time-varying Weibull parameters following [26], which are commonly used for wind characterization [14]. Fig. 1 illustrates that nearby wind farms exhibit stronger co-movements than distant ones.

Wind speeds are mapped to power outputs via a turbine power curve [27], and a point forecast $\hat{p}_{i,h}$ is constructed to represent the day-ahead forecast used for bidding. The forecast residual is defined by

$$r_{i,h} = p^{\text{true}}_{i,h} - \hat{p}_{i,h}, \tag{3}$$

and residual scenarios are generated with heavy-tailed marginals and the same cross-farm dependence structure, which yields $\{r_{i,h,s}\}$ for each hour h and scenario s. Power scenarios are obtained by

$$p_{i,h,s} = \hat{p}_{i,h} + r_{i,h,s}. \tag{4}$$

In the baseline setting, residual scenarios are generated independently for each hour, so temporal correlation across hours is not represented and tail risk under persistent low-wind events can be underestimated. Detailed fitting choices and diagnostics are reported in Appendix, and temporal extensions are discussed as future work.

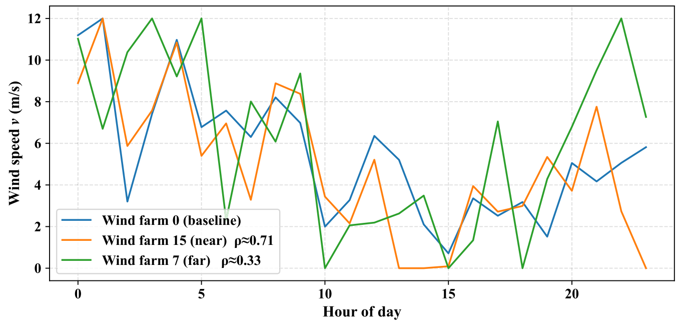

Fig. 1. Illustrative example of wind speed generation.

3. Joint optimization of grouping and aggregated bidding under uncertainty

This section develops a risk-aware mixed-integer programming framework that jointly determines group formation and aggregated day-ahead bidding strategies for multiple wind farms under scenario-based uncertainty. The formulation captures the uncertainty-reduction effect arising from grouping heterogeneous wind farms with complementary forecast errors, where the over- and under-generation characteristics and tail dependence among farms are exploited to mitigate joint imbalance risks.

In the present study, $\pi_{i,h}$ denotes the exogenous day-ahead price input for wind farm i at hour h, rather than an endogenous locational marginal price obtained from a network-constrained dispatch model. Aggregated bids are settled against portfolio imbalances at the group level, with lower-tail exposure managed through a CVaR-based penalty. Electricity prices are treated as exogenously given, node-specific day-ahead prices, and aggregated bids are settled against portfolio imbalances at the group level, with lower-tail exposure managed through a CVaR-based penalty. For benchmarking, an individual bidding model is also formulated as a linear programming baseline employing the same CVaR linearization. The optimization is formulated and solved for each operating hour h, and inter-hour couplings are omitted in the baseline model. This assumption is consistent with the hourly bidding interval typically used in day-ahead markets, which is considerably longer than the control timescale of individual wind turbines, whose output can flexibly adjust within each hour.

3.1. CVaR-based mixed-integer co-optimization

Using the notation defined in the Nomenclature, a CVaR-based mixed-integer co-optimization model is formulated for each operating hour h. Scenario realizations $p_{i,h,s}$ are constructed as in Section 2, so that both cross-farm dependence and marginal distributions are preserved. In this formulation, complementarity is reflected through the objective that penalizes the tail risk of aggregated imbalance losses, and it is further quantified and reported ex-post in Section 5.3 for verification.

Binary variables $z_{i,g} \in \{0,1\}$ indicate the assignment of wind farm i to candidate group g, and satisfy

$$\sum_{g\in\mathcal{G}} z_{i,g} = 1,\ \forall i \in \mathcal{I}, \tag{5}$$

so that each wind farm is assigned to exactly one group. In practice, not all wind farms can or will cooperate, due to ownership boundaries, contractual restrictions, market rules, or strategic rivalry. To represent these limitations, we define an incompatibility set $\mathcal{E} \subseteq \{(i,j) \mid i < j\}$. The set $\mathcal{E}$ is an exogenously specified input to the grouping model and is not intended to approximate electrical-network topology, transmission feasibility, or power-flow constraints. Rather, it is used only to represent institutional or strategic cooperation limits in the case study. For any $(i,j) \in \mathcal{E}$, wind farms i and j are not allowed to be assigned to the same group, which is enforced by

$$z_{i,g} + z_{j,g} \le 1,\ \forall (i,j) \in \mathcal{E},\ \forall g \in \mathcal{G}. \tag{6}$$

Given these assignments, each group $g \in \mathcal{G}$ submits a nonnegative day-ahead bid $Q_{g,h} \ge 0$ (MW). To incorporate nodal prices while preserving linearity, the group schedule is represented by nonnegative allocation variables $\theta_{i,g,h}$ (MW) that apportion $Q_{g,h}$ across nodes:

$$\sum_{i\in\mathcal{I}} \theta_{i,g,h} = Q_{g,h}, \forall g \in \mathcal{G},\ \forall h \in \mathcal{H}, \tag{7a}$$

$$0 \le \theta_{i,g,h} \le C_i\, z_{i,g}, \forall i \in \mathcal{I},\ \forall g \in \mathcal{G},\ \forall h \in \mathcal{H}. \tag{7b}$$

These constraints ensure that each group bid is fully allocated across its members and that the share assigned to each wind farm does not exceed its installed capacity.

If a capacity-weighted group price is desired for ex-post reporting, it is convenient to define

$$\bar{\pi}_{g,h} = \frac{\sum_{i\in\mathcal{I}} C_i\, \pi_{i,h}\, z_{i,g}}{\sum_{i\in\mathcal{I}} C_i\, z_{i,g}}, \tag{8}$$

which recovers the identity $\sum_i \pi_{i,h}\theta_{i,g,h} = \bar{\pi}_{g,h} Q_{g,h}$ whenever $\theta_{i,g,h}$ is proportional to $C_i\, z_{i,g}$. Eq. (8) is not used within the optimization but is useful for interpreting the resulting group bids.

Real group production in scenario $s \in \mathcal{S}$ is

$$P_{g,h,s} = \sum_{i\in\mathcal{I}} p_{i,h,s}\, z_{i,g}, \tag{9}$$

and group under- and over-production are captured by deviation variables satisfying

$$\xi^{\downarrow}_{g,h,s} \ge Q_{g,h} - P_{g,h,s}, \tag{10a}$$

$$\xi^{\uparrow}_{g,h,s} \ge P_{g,h,s} - Q_{g,h}, \tag{10b}$$

$$\xi^{\downarrow}_{g,h,s}, \xi^{\uparrow}_{g,h,s} \ge 0,\ \forall g \in \mathcal{G},\ \forall h \in \mathcal{H},\ \forall s \in \mathcal{S}, \tag{10c}$$

which define the magnitudes of group shortfall and surplus for each hour and scenario.

The per-scenario hourly profit of the entire portfolio is

$$\mathrm{Prof}_{h,s} = \sum_{g\in\mathcal{G}} \sum_{i\in\mathcal{I}} \pi_{i,h}\, \theta_{i,g,h} - \kappa^{\downarrow}_h \sum_{g\in\mathcal{G}} \xi^{\downarrow}_{g,h,s} - \kappa^{\uparrow}_h \sum_{g\in\mathcal{G}} \xi^{\uparrow}_{g,h,s}. \tag{11}$$

The corresponding loss is defined as $L_{h,s} = -\mathrm{Prof}_{h,s}$, and a CVaR penalty at confidence $\alpha \in (0,1)$ is introduced using the Rockafellar–Uryasev epigraph representation [21] with auxiliary variables $\eta_h \in \mathbb{R}$ and $u_{h,s} \ge 0$:

$$u_{h,s} \ge L_{h,s} - \eta_h,\ u_{h,s} \ge 0,\ \forall s \in \mathcal{S}. \tag{12}$$

The hourly objective maximizes expected profit penalized by CVaR and spatial dispersion:

$$\max \sum_{s\in\mathcal{S}} w_s\, \mathrm{Prof}_{h,s} - \lambda \left(\eta_h + \frac{1}{1-\alpha} \sum_{s\in\mathcal{S}} w_s\, u_{h,s} \right) \tag{13}$$

where w_s are scenario probabilities (for equal weighting $w_s = 1/|\mathcal{S}|$) and $\lambda \ge 0$ is the risk-aversion weight. In practice, under-generation is more critical for system security than over-generation, since shortages must be covered by reserves and may stress conventional units, whereas surpluses can be mitigated by curtailment. Accordingly, the penalty coefficients are chosen such that $\kappa^{\downarrow}_h > \kappa^{\uparrow}_h$, reflecting the higher operational risk of shortfalls.

Collecting the above elements, the co-bidding problem for a fixed hour h can be written compactly as

$$\begin{aligned} &\text{Maximize } (13) \\ &\qquad \text{s.t. } (5) - (37). \end{aligned} \tag{14}$$

This formulation is a mixed-integer linear program with binary variables $z_{i,g}$ for group membership and continuous variables for bids, allocations, imbalances, and CVaR terms. Since inter-hour coupling is not imposed, the problems for different hours are decoupled and can be solved independently and in parallel by standard MILP solvers. Moreover, inter-hour deliverability constraints such as ramping limits are not expected to be binding under an hourly schedule, because active-power controllability requirements for wind power plants are typically specified at sub-hour time scales, with representative ramp-rate limits commonly stated in MW/min or as fractions of rated power (e.g., on the order of 3%–10% of rated capacity per minute) [32]. The incompatibility constraints are linear and preserve the MILP structure.

3.2. Individual bidding case for benchmarking

For benchmarking, an individual bidding model is considered in which each farm i submits its own bid $q_{i,h} \geq 0$ (MW). The per-scenario profit is

$$\mathrm{Prof}^{\mathrm{ind}}_{h,s} = \sum_{i\in\mathcal{I}} \pi_{i,h}\, q_{i,h} - \kappa^{\downarrow}_h \sum_{i\in\mathcal{I}} \xi^{\downarrow}_{i,h,s} - \kappa^{\uparrow}_h \sum_{i\in\mathcal{I}} \xi^{\uparrow}_{i,h,s}, \tag{15}$$

subject to

$$0 \leq q_{i,h} \leq C_i, \tag{16a}$$

$$\xi^{\downarrow}_{i,h,s} \geq q_{i,h} - p_{i,h,s}, \tag{16b}$$

$$\xi^{\uparrow}_{i,h,s} \geq p_{i,h,s} - q_{i,h}, \tag{16c}$$

$$\xi^{\downarrow}_{i,h,s}, \xi^{\uparrow}_{i,h,s} \geq 0,\ \forall i \in \mathcal{I},\ \forall s \in \mathcal{S}. \tag{16d}$$

The same hourly CVaR epigraph (12) is used with $L_{h,s} = -\mathrm{Prof}^{\mathrm{ind}}_{h,s}$, and the objective adopts the form of (13) with group-related terms removed. This formulation reduces to a linear program and serves as a natural benchmark, since it preserves the same prices, penalties, scenario set, and CVaR treatment as in the aggregated case, differing only by the exclusion of grouping variables. Both optimization problems can be solved by commercial solvers such as CPLEX or Gurobi.

4. Settlement within aggregated bidding groups

This section develops a settlement mechanism for aggregated day-ahead bidding that allocates energy revenue and the benefits from risk reduction in a transparent and budget-balanced manner. The settlement design is kept consistent with the risk-aware objective used in the grouping-and-bidding co-optimization model. This settlement layer is a central part of the proposed framework, because the economic value created by uncertainty complementarity becomes practically meaningful only when it can be redistributed among group members through a transparent and internally consistent rule. Under aggregated bidding, two types of revenues need to be allocated within each group: (i) energy revenue from the group bid cleared at the market price, and (ii) the reduction in extreme imbalance penalties. To avoid double-counting and to align incentives, the energy revenue is first allocated in a simple capacity-proportional manner, and the risk-savings pool is then distributed according to responsibility measured on CVaR-determining tail scenarios. The former provides a stable and transparent settlement of deterministic energy value, while the latter redistributes uncertainty-reduction benefits according to tail-risk responsibility under realized scenarios.

4.1. Allocation of group energy revenue

Aggregation can change the wind farms that participate in market bidding, but it does not alter how energy is settled at the cleared market price. To maintain incentive transparency and to separate energy and risk effects, the group's energy revenue should be allocated purely in proportion to capacity shares. The capacity-weighted group price $\bar{\pi}_{g,h}$ is defined as in (8). Then, the per-hour energy revenue of group g and its member-level allocation are given by:

$$R^{\mathrm{energy}}_{g,h} = \bar{\pi}_{g,h} Q_{g,h}, \tag{17a}$$

$$r^{\mathrm{energy}}_{i,h} = R^{\mathrm{energy}}_{g,h} \frac{C_i}{\sum_{j\in g} C_j}, \tag{17b}$$

$$r^{\mathrm{energy,day}}_{i} = \sum_{h} r^{\mathrm{energy}}_{i,h}. \tag{17c}$$

This allocation rule deliberately keeps the energy revenue independent of risk-savings and tail responsibilities, ensuring that energy revenue sharing remains stable, predictable, and depends only on $\bar{\pi}_{g,h}, Q_{g,h}, C_i$, while the subsequent risk-savings allocation focuses solely on the uncertainty-reduction effect. A capacity-proportional rule is adopted here because installed capacity reflects each member's committed asset scale in the aggregated portfolio. This makes the deterministic energy settlement transparent, stable, and easy to verify, and each participant can anticipate its energy-revenue share ex ante.

4.2. Allocation of uncertainty reduction benefits

In what follows, only the reduction in group-level risk achieved by aggregation, relative to individual bidding, is quantified and redistributed in a budget-balanced way. A tail-consistent allocation rule based on the RU-CVaR objective is adopted to decompose total risk savings into member-specific contributions via tail-responsibility weights, so that payouts are tied to the scenarios that drive aggregate tail risk while preserving internal fairness across participants.

4.2.1. Group and individual losses under CVaR

For comparability of losses, the Group-level shortfall and surplus for a fixed operating hour h and scenarios $s \in \mathcal{S}$ can be defined as:

$$\Delta U_{g,h,s} = \max\left\{\left(Q_{g,h} - \textstyle\sum_i p_{i,h,s} z_{i,g}\right), 0\right\}, \tag{18a}$$

$$\Delta O_{g,h,s} = \max\left\{\left(\textstyle\sum_i p_{i,h,s} z_{i,g} - Q_{g,h}\right), 0\right\}. \tag{18b}$$

Then, the corresponding induced losses are defined from imbalance in group power generation as:

$$L_{g,h,s} = \kappa^{\downarrow}_h \Delta U_{g,h,s} + \kappa^{\uparrow}_h \Delta O_{g,h,s}. \tag{19}$$

For benchmarking, the individual-bidding loss is

$$L^{\mathrm{ind}}_{i,h,s} = \kappa^{\downarrow}_h \max\left\{\left(q_{i,h} - p_{i,h,s}\right), 0\right\} + \kappa^{\uparrow}_h \max\left\{\left(p_{i,h,s} - q_{i,h}\right), 0\right\}. \tag{20}$$

All CVaR terms use the same confidence level α and the same penalties $\kappa^{\downarrow}_h/\kappa^{\uparrow}_h$ so that group and individual risks are strictly comparable.

To compare group and individual risks consistently and to evaluate diversification, the Rockafellar–Uryasev (RU) formulation of CVaR is adopted, which is also the risk measure used in the bidding model. For completeness, the epigraph representation is recalled:

$$\begin{aligned}&\mathrm{CVaR}_\alpha\left(\{L_{g,h,s}\}_s\right) = \tau_{g,h} + \frac{1}{(1-\alpha)} \sum_{s\in\mathcal{S}} w_s\, u_{g,h,s},\\ &u_{g,h,s} \geq L_{g,h,s} - \tau_{g,h},\quad u_{g,h,s} \geq 0,\end{aligned} \tag{21}$$

where $\{L_{g,h,s}\}_s$ denotes the finite collection of loss realizations indexed by $s \in \mathcal{S}$. The induced tail set is then defined as

$$T_{g,h} = \left\{s \in \mathcal{S} \mid L_{g,h,s} \geq \tau_{g,h}\right\}. \tag{22}$$

The following proposition is introduced to identify which scenarios belong to the empirical tail that determines CVaR_α within this formulation. It will be used to define scenario-level responsibilities that are consistent with the risk objective.

Remark 1 (Tail-set characterization under the Rockafellar–Uryasev CVaR formulation). At an optimal solution of (21) with equal weights $w_s = 1/|\mathcal{S}|$, it holds that $u_{g,h,s} = \max\{L_{g,h,s} - \tau_{g,h}, 0\}$ for all s. Moreover, $T_{g,h} = \{s \mid u_{g,h,s} > 0\}$ coincides with the empirical $(1-\alpha)$-tail that determines CVaR_α.

This result implies that only scenarios in the tail set $T_{g,h}$ influence CVaR_α. In what follows, responsibility is therefore assessed on $T_{g,h}$, and group imbalances are decomposed across members at the scenario level.

4.2.2. Scenario-level imbalance attribution

To construct this attribution, capacity-proportional baseline shares are first introduced:

$$B_{i,h}^{\text{share}} = Q_{g,h} \frac{C_i}{\sum_{j \in g} C_j}, \tag{23}$$

Here, $B_{i,h}^{\text{share}}$ is used only as an ex-ante benchmark for scenario-level imbalance attribution, rather than as the final basis for redistributing group benefits. Since the group bid $Q_{g,h}$ is itself an ex-ante commitment, the reference used to decompose realized group imbalance should also be fixed ex ante. Installed capacity is observable, easy to verify, and does not depend on realized output or realized forecast errors. For this reason, it is used here as a stable accounting reference. By contrast, a benchmark based on realized output would depend on the same uncertainty that the settlement rule is trying to allocate, while a forecast-dependent benchmark could create additional room for manipulation. Under this benchmark, member deviations are measured separately in the under-production and over-production directions, so that responsibility is assessed relative to a common ex-ante reference and opposite-direction deviations are not mixed in the same attribution step. The associated under- and over-production indicators are defined as

$$a_{i,h,s} = \max\left\{ \left(B_{i,h}^{\text{share}} - p_{i,h,s} \right), 0 \right\}, \tag{24a}$$

$$b_{i,h,s} = \max\left\{ \left(p_{i,h,s} - B_{i,h}^{\text{share}} \right), 0 \right\}. \tag{24b}$$

Group imbalances are then allocated proportionally to these contributions:

$$\delta U_{i,h,s} = \Delta U_{g,h,s} \frac{a_{i,h,s}}{\sum_{j \in g} a_{j,h,s}}, \tag{25a}$$

$$\delta O_{i,h,s} = \Delta O_{g,h,s} \frac{b_{i,h,s}}{\sum_{j \in g} b_{j,h,s}}. \tag{25b}$$

This proportional allocation links member-level attribution directly to same-direction deviations from the fixed benchmark. As a result, the attributed under- and over-production exactly sum to the realized group under- and over-production in each scenario, so that no imbalance quantity is lost or artificially created during attribution. In addition, for a fixed realized group imbalance, a member with a larger deviation from the benchmark in one direction receives a larger attributed imbalance in the same direction.

Given $\delta U_{i,h,s}$ and $\delta O_{i,h,s}$, the member-level loss contribution in scenario s is defined as

$$L_{i,h,s}^{\text{contrib}} = \kappa_h^{\downarrow} \delta U_{i,h,s} + \kappa_h^{\uparrow} \delta O_{i,h,s}. \tag{26}$$

These definitions provide a scenario-level attribution of group losses to individual members, where each contribution reflects both the magnitude and the direction of the deviation from the baseline. The scenario dependence of this attribution is therefore not arbitrary. It arises because imbalance penalties are realized only after uncertainty is realized, so the decomposition of group imbalance must also be based on realized outcomes. At the same time, the benchmark $B_{i,h}^{\text{share}}$, the attribution logic, and the loss definition remain the same across scenarios. In this sense, only the realized deviations vary across scenarios, while the attribution principle itself remains unchanged.

The member-level loss contribution $L_{i,h,s}^{\text{contrib}}$ uses the same imbalance-penalty coefficients as the group loss definition, so the attribution remains consistent with the loss model used in the bidding problem. The separate question of numerical stability under finite scenario sampling is examined in the supplementary scenario-size analysis in Section 5.5.

Proposition 1 (Conservation and monotonicity of imbalance attribution). *For any scenario s, $\sum_{i \in g} \delta U_{i,h,s} = \Delta U_{g,h,s}$ and $\sum_{i \in g} \delta O_{i,h,s} = \Delta O_{g,h,s}$. Moreover, for fixed $\Delta U_{g,h,s}$ and $\{a_{j,h,s}\}_{j \neq i}$, $\delta U_{i,h,s}$ is strictly increasing in $a_{i,h,s}$. Similarly, for fixed $\Delta O_{g,h,s}$ and $\{b_{j,h,s}\}_{j \neq i}$, $\delta O_{i,h,s}$ is strictly increasing in $b_{i,h,s}$.*

These properties ensure that the scenario-level attributions are internally consistent, meaning that group imbalances are exactly decomposed into member-level components, and members with larger deviations assume proportionally greater scenario losses. The proof is given in Appendix.

4.2.3. Tail-responsibility weights and risk-savings allocation

Building on this consistency, the scenario losses are aggregated over the tail realizations relevant to CVaR_α, thereby defining the corresponding tail-responsibility weights as follows:

$$\omega_{i,g,h} = \frac{\sum_{s \in T_{g,h}} L_{i,h,s}^{\text{contrib}}}{\sum_{s \in T_{g,h}} L_{g,h,s}}, \quad \sum_{i \in g} \omega_{i,g,h} = 1. \tag{27}$$

It quantifies each member's share of the aggregate loss within the tail set $T_{g,h}$. The weights are evaluated precisely on the scenarios that drive CVaR_α, thereby establishing a direct and coherent link between individual responsibilities and the overall risk objective. Although the rule is expressed on a finite scenario set for consistency with the CVaR-based bidding model, its practical implementation does not require hypothetical scenarios at settlement time. Once outputs are realized, the same attribution formulas can be applied directly to realized deviations, and the corresponding empirical realizations over the settlement horizon can be used to identify the relevant tail outcomes for ex-post redistribution.

Extending the member-level attribution to the group perspective, the effect of aggregation on tail risk can be represented by the per-hour risk-savings pool,

$$RS_{g,h} = \sum_{i \in g} \text{CVaR}_\alpha \left(\{ L_{i,h,s}^{\text{ind}} \}_s \right) - \text{CVaR}_\alpha \left(\{ L_{g,h,s} \}_s \right) \geq 0, \tag{28}$$

which measures the benefit from uncertainty reduction achieved through aggregation. Groups that realize larger improvements in tail performance thus generate higher $RS_{g,h}$, reflecting their group-level gain from risk diversification in stochastic outcomes. This group-level benefit is then allocated to individual members according to the previously defined responsibility weights:

$$r_{i,h}^{\text{risk}} = \omega_{i,g,h} RS_{g,h}. \tag{29}$$

To ensure that this construction behaves coherently, the following property is established.

Proposition 2 (Nonnegativity of $RS_{g,h}$ under a CVaR dominance requirement). *Let $\{q_{i,h}\}_{i \in g}$ be any set of individual nominations and $Q_{g,h}^{bench} = \sum_{i \in g} q_{i,h}$. If the adopted group bid $Q_{g,h}$ weakly dominates this benchmark in CVaR at level α, then $RS_{g,h} \geq 0$. Moreover, the allocation is budget-balanced: $\sum_{i \in g} r_{i,h}^{risk} = RS_{g,h}$.*

Under the dominance condition, aggregation cannot generate a negative risk-savings pool. Allocating $RS_{g,h}$ in proportion to the tail-responsibility weights preserves budget balance by construction. Consequently, members who bear a larger share of the attributed tail losses receive a proportionally larger share of the risk-savings pool, so that the redistribution is aligned with residual tail exposure while preserving budget balance.

To further clarify the direction of this payment rule, define

$$x_{i,g,h} = \sum_{s \in T_{g,h}} L_{i,h,s}^{\text{contrib}}, \; S_{g,h} = \sum_{j \in g} x_{j,g,h} = \sum_{s \in T_{g,h}} L_{g,h,s},$$

where the last equality follows from Proposition 1. Then $\omega_{i,g,h} = x_{i,g,h}/S_{g,h}$. The following result is not a full game-theoretic incentive claim. It only shows a local monotonicity property of the risk-savings payment when the other members' tail-loss contributions are fixed.

Proposition 3 (**Local monotonicity of the risk-savings payment**). *Fix a group* g *and an operating hour* h. *If the other members' tail-loss contributions are fixed and*

$$\frac{\partial RS_{g,h}}{\partial x_{i,g,h}} \le -\frac{S_{g,h} - x_{i,g,h}}{x_{i,g,h} S_{g,h}} RS_{g,h},$$

then $r^{\text{risk}}_{i,h}$ *is non-increasing in* $x_{i,g,h}$. *Therefore, under this condition, member* i *cannot increase its own risk-savings payment by increasing its attributed tail-loss contribution.*

This result clarifies the intended direction of the payment rule. A larger tail-loss contribution raises the member's responsibility weight, but it can also reduce the total risk-savings pool. When the latter effect is sufficiently strong, a larger attributed tail-loss contribution does not lead to a larger payment. This result should be read as a supporting property of the settlement rule in the cooperative setting considered here, rather than as a full guarantee against all strategic behavior. The proof is given in Appendix B.3.

4.2.4. Member-level daily settlement

Finally, energy revenues and risk-savings benefits are combined to determine each member's daily settlement. The total payment for member i aggregates its share of deterministic energy outcomes and its allocation of tail-risk reduction:

$$R^{\text{tot}}_i = r^{\text{energy,day}}_i + \sum_h r^{\text{risk}}_{i,h}. \tag{30}$$

The two components are additively separated, ensuring that the benefit from risk reduction is not double-counted and that the incentives remain transparent. Capacity primarily determines each member's energy-sharing outcome, while its tail responsibility governs the distribution of the group-level risk-savings. This separation clarifies how energy revenue and risk contribution jointly shape the overall settlement and reinforces consistency between physical participation and financial rewards, which will be further examined through empirical fairness and efficiency metrics in the next subsection.

4.3. Comparative allocation rules and evaluation metrics

To highlight the fairness and risk-sharing properties of the proposed tail-consistent settlement, several alternative allocation rules are introduced together with a set of metrics that visualize their distinctive properties. In all cases, energy revenue is allocated in proportion to installed capacity as in (17), while only the collective risk-savings pool $RS_{g,h}$ in (28) is redistributed among members.

4.3.1. Other allocation strategies for comparison

For comparison, several benchmark allocation rules are written in a common form. In each case, the risk-savings allocation is represented through normalized weights $\{\psi_{i,h}\}_{i\in g}$ such that

$$r^{\text{risk}}_{i,h} = \psi_{i,h} RS_{g,h}, \ \sum_{i\in g} \psi_{i,h} = 1, \ \psi_{i,h} \ge 0, \tag{31}$$

where differences across rules are entirely reflected by the definition of $\psi_{i,h}$.

(1) Tail-consistent rule.

For the proposed tail-consistent rule, the weights coincide with the tail-responsibility coefficients:

$$\psi^{\text{tail}}_{i,h} = \omega_{i,g,h}, \tag{32}$$

with $\omega_{i,g,h}$ defined in (27). For alignment evaluation, it is convenient to recall the normalized tail-loss contributions constructed from (26) on the RU-CVaR tail set (22):

$$c_{i,h} = \frac{\sum_{s\in T_{g,h}} L^{\text{contrib}}_{i,h,s}}{\sum_{j\in g}\sum_{s\in T_{g,h}} L^{\text{contrib}}_{j,h,s}}, \tag{33}$$

which will serve as the reference in the alignment metric below.

(2) All-scenarios rule.

To test sensitivity to non-tail attribution, an all-scenarios rule replaces the tail set $T_{g,h}$ with the entire scenario set S:

$$\psi^{\text{allscen}}_{i,h} = \frac{\sum_{s\in S} L^{\text{contrib}}_{i,h,s}}{\sum_{j\in g}\sum_{s\in S} L^{\text{contrib}}_{j,h,s}}. \tag{34}$$

(3) Capacity-based rule.

A first proportional baseline relates payouts directly to installed capacity:

$$\psi^{\text{cap}}_{i,h} = \frac{C_i}{\sum_{j\in g} C_j}. \tag{35}$$

(4) Energy-based rule.

A second proportional baseline relates payouts to expected energy production:

$$E_{i,h} = \sum_{s\in S} |S|^{-1} p_{i,h,s}, \tag{36a}$$

$$\psi^{\text{eng}}_{i,h} = \frac{E_{i,h}}{\sum_{j\in g} E_{j,h}}, \tag{36b}$$

where $E_{i,h}$ denotes the scenario-weighted expected energy production of member i at hour h.

(5) Equal-share rule.

An equal-share baseline assigns uniform weights irrespective of size and exposure:

$$\psi^{\text{eq}}_{i,h} = \frac{1}{|g|}. \tag{37}$$

(6) Shapley-based rule.

Finally, a Shapley-based benchmark is computed from the coalition worth defined by risk-savings:

$$v_h(S) = \sum_{i\in S} \text{CVaR}_\alpha(\{L^{\text{ind}}_{i,h,s}\}_s) - \text{CVaR}_\alpha(\{L_{S,h,s}\}_s), \tag{38}$$

with the Shapley value

$$\phi^{\text{Sh}}_{i,h} = \sum_{S\subseteq N\setminus\{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} \left(v_h(S\cup\{i\}) - v_h(S)\right), \tag{39}$$

and normalized contribution to the risk-savings pool,

$$\psi^{\text{shapley}}_{i,h} = \frac{\phi^{\text{Sh}}_{i,h}}{\max\{RS_{g,h}, \varepsilon\}}, \ \varepsilon > 0, \tag{40}$$

where ε is a small positive constant (e.g., 10^{-9}) used only to avoid division by zero and has no effect when denominators are strictly positive.

4.3.2. Evaluation metrics

Building on the above allocation rules, the empirical comparison employs a set of metrics that capture four complementary properties, namely alignment with tail risk, individual-level protection, equity of per-MW payouts, and overall dispersion. These metrics enable a direct and consistent comparison of the proposed tail-consistent rule with the all-scenarios, capacity, energy, equal, and Shapley benchmarks.

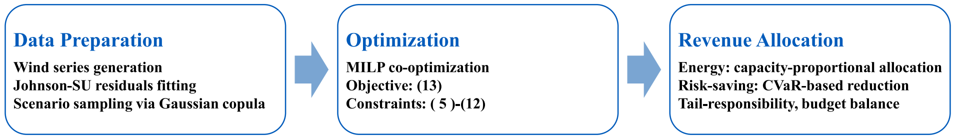

Fig. 2. The overall workflow of the case study.

(1) Tail-Alignment Coefficient (TAC).

Alignment with the RU-CVaR objective is measured through the Tail-Alignment Coefficient (TAC), which compares the rule's scenario weights $\{\psi_{i,h}\}$ with the reference tail weights $\{c_{i,h}\}$ defined in (33):

$$\mathrm{TAC}_h = 1 - \frac{1}{2}\sum_{i\in g} |\psi_{i,h} - c_{i,h}| \in [0,1], \tag{41}$$

where a value of one indicates perfect alignment with the tail-risk profile, while lower values reflect greater deviation. Higher scores therefore correspond to allocations that are more consistent with the scenarios determining CVaR_α.

(2) Individual-Risk Minimum (IRmin).

To further evaluate how risks are distributed within the group, the Individual-Risk Minimum (IRmin_h) is introduced to capture the minimum per-MW risk margin among members [19]:

$$\mathrm{IRmin}_h = \min_{i\in g} \frac{r^{\mathrm{risk}}_{i,h}}{C_i}, \tag{42}$$

where higher values denote stronger protection against negative risk exposure for the least-advantaged participant. While IRmin_h focuses on the most exposed member, assessing overall fairness requires examining how payoffs are distributed across all participants.

(3) Maximum Envy-Freeness Gap (EFmax).

To ensure comparability irrespective of participant size, payouts are normalized by capacity as per-MW values $u_{i,h} = r^{\mathrm{risk}}_{i,h}/C_i$. Differences in these normalized payouts capture distributional disparities and are summarized first by the Maximum Envy-Freeness Gap:

$$\mathrm{EFmax}_h = \max_{i,j\in g} \max\{0,\, u_{j,h} - u_{i,h}\}, \tag{43}$$

which measures the largest observed instance of ex-post envy between any two participants.

(4) Normalized Envy-Freeness Ratio (NEF).

A complementary indicator is the Normalized Envy-Freeness Ratio [11],

$$\mathrm{NEF}_h = \frac{1}{|g|(|g|-1)} \sum_{i\neq j} \mathbf{1}\{u_{j,h} > u_{i,h}\}, \tag{44}$$

which quantifies the share of unequal pairwise outcomes. Smaller values of EFmax_h and NEF_h imply fewer perceived inequities and more uniform treatment after normalizing for capacity.

(5) Gini coefficient (Gini).

Residual inequality in the overall allocation of payouts is captured by the Gini coefficient [18],

$$\mathrm{Gini}_h = \frac{\sum_{i\in g}\sum_{j\in g} |u_{i,h} - u_{j,h}|}{2\,|g|\,\max\{\sum_{i\in g} u_{i,h},\, \varepsilon\}},\ \varepsilon > 0, \tag{45}$$

where smaller values indicate a more balanced distribution of rewards per MW across the group.

Taken together, these metrics provide a comprehensive characterization of allocation outcomes. TAC_h captures adherence to the intended tail-risk profile, IRmin_h reflects the minimum level of protection among participants, and EFmax_h, NEF_h, and Gini_h collectively describe the degree of dispersion and fairness in the resulting payouts.

5. Case study

To verify the superiority and practical effectiveness of the proposed co-optimization framework, this case study evaluates the overall architecture from the perspectives of grouping, bidding, and intra-group revenue allocation. Simulated data are used in this study, generated according to the statistical characteristics and procedures described in Section 2. Real operational data are not employed because large-scale wind-farm datasets are generally confidential and unavailable for public research access. After constructing the wind-farm dataset, the grouping and bidding co-optimization problems are formulated and solved following the procedures outlined in Section 3. Subsequently, the intra-group revenue allocation for each member is conducted according to the settlement rules presented in Section 4, and the overall experimental workflow is illustrated in Fig. 2.

5.1. Basic setup

The co-optimization study is conducted on a system with 20 wind farms. Their geographical coordinates and installed capacities are randomly sampled within prescribed bounds, and historical production data from the most recent 30 days are used to generate wind power scenarios. The CVaR confidence level is fixed at $\alpha = 0.95$. Site-specific exogenous day-ahead price profiles are used to represent market-price heterogeneity in the case study. These price profiles are synthetically generated from a time-of-use template rather than from a network or dispatch model. Specifically, a common daily TOU structure is first defined by a base price, peak-hour increments, and off-peak reductions. The wind farms are then partitioned into several price zones according to their spatial coordinates, and zone-dependent multipliers are applied to create spatial heterogeneity. A mild spatial gradient and small site-specific perturbations are further added, so that different wind farms face similar but non-identical exogenous day-ahead price trajectories. Therefore, the price inputs used in the simulations are treated as known external day-ahead prices, and no congestion or power-flow verification is performed in the present case study. As shown in Fig. 3, the four example wind farms connected to nodes 13, 14, 16, and 18 exhibit weekday-like diurnal price trajectories with two pronounced peaks: one in the morning when industrial and commercial activities begin, and another in the evening when residential demand rises. Prices remain elevated during daytime and decrease overnight as system demand subsides.

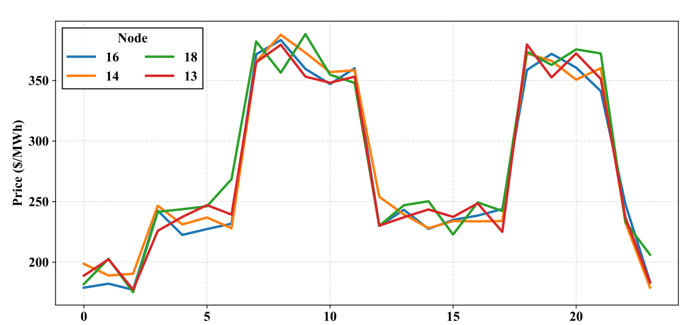

Fig. 3. Hourly exogenous day-ahead price profiles for four representative wind farms on a typical weekday.

Key parameter choices are as follows. To reflect practical participation limits in cooperative bidding, we introduce an incompatibility set $\mathcal{E}$ in (6), where any $(i,j) \in \mathcal{E}$ indicates that wind farms i and j cannot be assigned to the same group due to ownership boundaries, contractual restrictions, or strategic rivalry. In this case study, $\mathcal{E}$ is generated synthetically to represent these non-technical constraints, and the resulting grouping decisions are required to satisfy (6). The penalties for over-generation and under-generation are set to $\kappa_h^{\uparrow} = 50$ and $\kappa_h^{\downarrow} = 200$, respectively, to reflect grid safety and reliability concerns [35]. These values define the baseline case for the numerical experiments, and all CVaR-based quantities are computed from the finite scenario set.

5.1.1. Adequacy of scenario sampling for CVaR-based quantities

The tail set $T_{g,h}$, the weights $\omega_{i,g,h}$, and the risk-savings pool $RS_{g,h}$ are computed from a finite number of scenarios via (21). Since these CVaR-based quantities enter the allocation and comparison results in this section, it is useful to check that the chosen scenario sample size S yields sufficiently accurate estimates. The following result summarizes standard finite-sample guarantees for the SAA estimator of CVaR.

Proposition 4 (Finite-sample accuracy of SAA–CVaR). *If the α-quantile is unique and the loss density is bounded away from zero near it, then for any $\delta \in (0,1)$,*

$$\mathbb{P}\left(\left|\widehat{CVaR}_{\alpha,S} - CVaR_\alpha\right| \le \frac{B}{1-\alpha}\sqrt{\frac{2\log(2/\delta)}{S}} + \frac{c}{(1-\alpha)\sqrt{S}}\right) \ge 1-\delta, \tag{46}$$

and

$$\sqrt{S}\left(\widehat{CVaR}_{\alpha,S} - CVaR_\alpha\right) \Rightarrow \mathcal{N}\left(0,\ \frac{Var\left((L-v_\alpha)_+\right)}{(1-\alpha)^2}\right). \tag{47}$$

This proposition shows that the sampling error of the SAA estimator for CVaR decreases at rate of $O(S^{-1/2})$ and admits an asymptotically normal limit. These bounds indicate that using a finite scenario set in the case study is reasonable and that the CVaR-based quantities $RS_{g,h}$ and $\omega_{i,g,h}$ can be treated as reliable numerical approximations of the corresponding true values for the scenario sizes used. The proof is given in Appendix.

5.2. Wind farms grouping results

The grouping result of the 20 wind farms at $h = 16$ is presented in Fig. 4. The groups are obtained by solving the proposed co-optimization model subject to the incompatibility constraints (6). These constraints represent practical participation limits, such as ownership boundaries, contractual restrictions, or strategic rivalry, and ensure that wind farms in any incompatible pair $(i,j) \in \mathcal{E}$ are never assigned to the same group. In Fig. 4, the gray dashed lines indicate the incompatible wind-farm pairs in $\mathcal{E}$, which are excluded from being grouped together. Under these feasibility constraints, the risk-aware objective still promotes grouping patterns that reduce aggregated imbalance losses and their tail exposure, thereby improving the robustness of market participation.

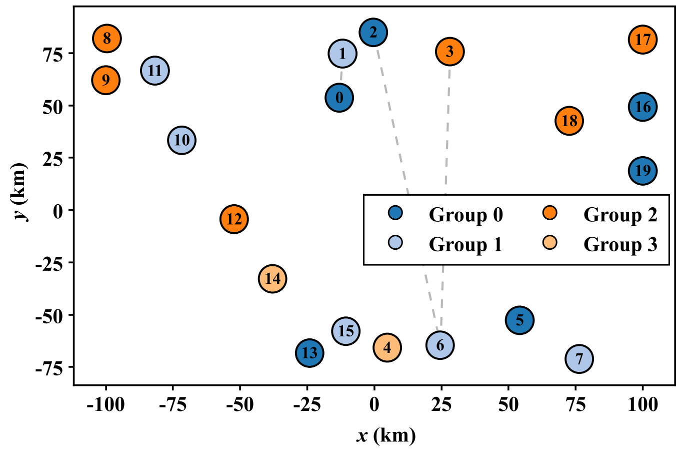

Fig. 4. Grouping results of 20 wind farms at $h = 16$ under the incompatibility constraints (6), where gray dashed lines indicate incompatible wind-farm pairs in $\mathcal{E}$.

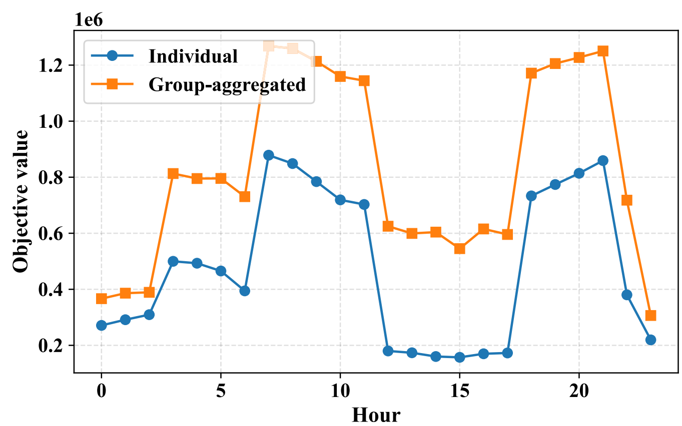

Fig. 5. Comparison of hourly risk-adjusted objective values between the grouped-aggregated and individual-bidding schemes.

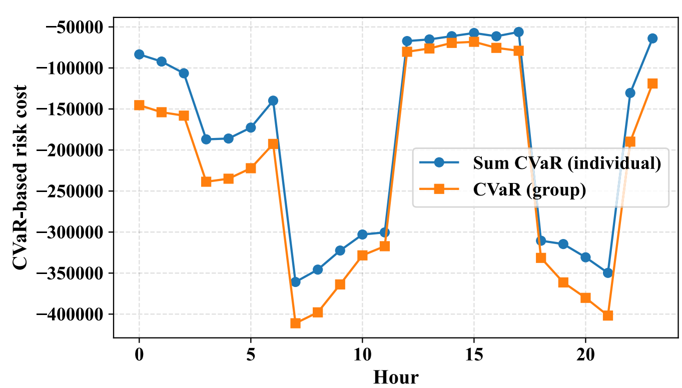

Fig. 6. Hourly comparison of CVaR-based risk costs between the grouped-aggregated and individual-bidding schemes.

5.3. Comparison of group-aggregated bidding strategy and individual bidding strategy

To highlight the advantage of the proposed grouping–bidding co-optimization framework, a comparison is first conducted between the hourly risk-adjusted objective values obtained from the group-aggregated scheme and those from the individual-bidding benchmark. As illustrated in Fig. 5, the objective values achieved by the grouped scheme are consistently higher, indicating that the proposed framework effectively enhances market performance by aggregating the temporal and spatial uncertainties of multiple wind farms. Through mutual compensation of forecast errors, the risks of over- and under-generation are mitigated, which in turn reduces deviation penalties and increases overall revenue.

Beyond the improvement in expected revenue, the proposed framework delivers a pronounced reduction in tail-risk exposure. As depicted in Fig. 6, the grouped-aggregated scheme consistently attains lower CVaR values across most hours compared with the individual bidding case. Although uncertainty complementarity is not imposed as an explicit term in the objective or constraints, it is promoted through

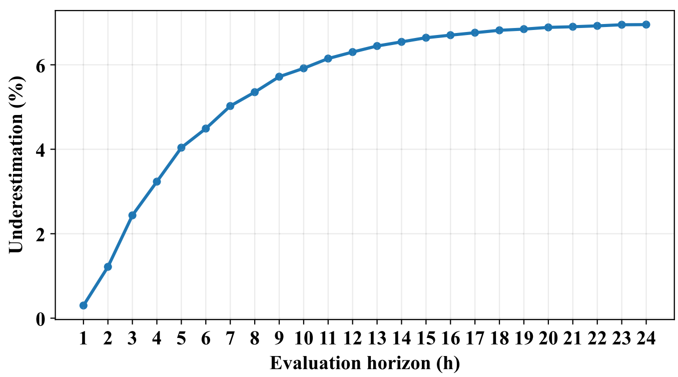

Fig. 7. Cumulative CVaR underestimation with temporal correlation.

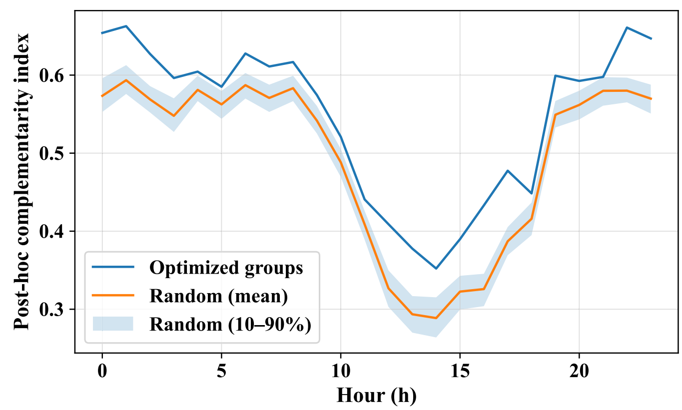

Fig. 8. Hourly ex-post tail complementarity index for the optimized grouping and a random grouping baseline.

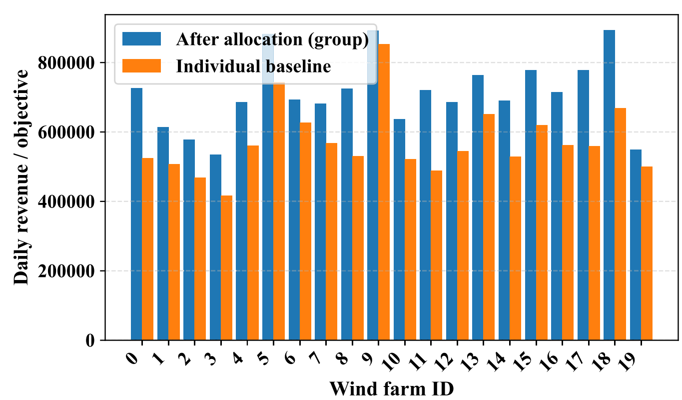

Fig. 9. Revenue allocation across 20 wind farms under the budget-balanced settlement.

the CVaR-based downside-loss penalization, i.e., grouping decisions that yield stronger mutual compensation of unfavorable deviations are preferentially selected because they reduce tail losses. Consequently, the joint optimization reduces overall portfolio uncertainty and enhances the robustness of market participation under variable renewable conditions. Since the baseline scenarios are generated independently for each hour, the impact of ignoring temporal correlation on CVaR estimation is quantified below.

A supplementary test is conducted to quantify the potential underestimation of tail risk caused by the hour-independent residual assumption. Temporally correlated residual trajectories are constructed by introducing positive short-lag dependence while preserving the same hourly marginals and the same cross-farm dependence. The cumulative multi-hour loss is then re-evaluated over an expanding horizon under the same confidence level. As shown in Fig. 7, the underestimation of cumulative CVaR is close to zero at very short horizons, then increases as adjacent unfavorable deviations become more likely to cluster in time, and finally approaches a stable range of about 6%–7% at longer horizons. This pattern is consistent with the short-range temporal dependence considered in the auxiliary test, where the incremental effect of temporal correlation decreases as the lag becomes larger. Therefore, the hour-independent setting provides a mildly optimistic estimate of cumulative tail exposure, while the qualitative comparison between grouped-aggregated bidding and individual bidding remains unchanged.

In addition to the CVaR comparison above, the induced uncertainty compensation within groups is verified by the following ex-post index.

To make the above mechanism verifiable, an ex-post tail complementarity index is reported, which is computed from the same scenario set and the optimized group membership. The index is defined in three steps. First, for each wind farm i and hour h, the scenario deviation from its expected production is formed as $e_{i,h,s} = p_{i,h,s} - \sum_{s' \in S} w_{s'} \, p_{i,h,s'}$. Second, for each group g, deviations are aggregated according to the optimized membership, $e_{g,h,s} = \sum_{i \in \mathcal{I}} z_{i,g} \, e_{i,h,s}$, and the downside deviation is extracted as $d^-_{(\cdot),h,s} = \max\{0, -e_{(\cdot),h,s}\}$. Third, the tail mean of downside deviations is evaluated at confidence level α, and the complementarity index is defined as

$$\mathrm{CI}^{\mathrm{tail}}_{g,h} = 1 - \frac{\mathrm{TailMean}_\alpha\left(d^-_{g,h,\cdot}\right)}{\sum_{i:z_{i,g}=1} \mathrm{TailMean}_\alpha\left(d^-_{i,h,\cdot}\right)}. \tag{48}$$

This ratio compares the group-level tail shortfall with the sum of members' tail shortfalls when treated separately. When little compensation is present, the ratio is close to one and $\mathrm{CI}^{\mathrm{tail}}_{g,h}$ is close to zero; when strong compensation is present, the group tail shortfall becomes much smaller than the sum and $\mathrm{CI}^{\mathrm{tail}}_{g,h}$ increases. In Fig. 8, the optimized grouping is compared with a random grouping baseline that preserves the same group-size distribution. The optimized results are consistently higher than the random baseline and are observed to exceed the upper band of random grouping in most hours, indicating that the achieved complementarity cannot be attributed to random aggregation. Therefore, uncertainty complementarity is shown to be effectively reflected by the optimized membership decisions, even though it is induced indirectly through the CVaR-based objective.

While the aggregated bidding strategy improves both the expected market performance and the portfolio's risk profile, it is also important to examine how these gains are distributed among individual participants, as the incentive for aggregation ultimately depends on member-level outcomes. As illustrated in Fig. 9, the ex-post settlement derived from the proposed budget-balanced rule achieves a more equitable distribution of revenues across the 20 wind farms compared with the individual-bidding benchmark, and the individual rationality condition is satisfied for all participants, with the vast majority of members obtaining higher revenues under aggregation than under their individual bids. This outcome can be attributed to the portfolio diversification effect, whereby temporal and spatial variations in wind generation partially offset one another. As a result, the aggregated forecast uncertainty is lowered, leading to smaller imbalance penalties and enabling a more stable and fair redistribution of market revenues. Overall, the results demonstrate that the efficiency and risk-reduction benefits achieved through the grouping–bidding co-optimization framework are effectively transmitted to the individual level without cross-subsidization, ensuring fairness, transparency, and consistent participation incentives.

5.4. Comparison of tail-consistent allocation strategy and other allocation rules

To evaluate the fairness and consistency of the proposed allocation mechanism, the tail-consistent rule is compared against five representative alternatives: all-scenarios, capacity-based, energy-based,

Table 2
Comparison of allocation metrics among different rules.

Allocation Rule	TAC ↑	IRmin ↑	EFmax ↓	Gini ↓
Tail-consistent	**1.000**	**10.109**	**54.681**	**0.069**
All-scenarios	0.991	10.478	54.772	0.070
Capacity-based	0.957	12.435	57.021	0.072
Energy-based	0.908	10.702	58.764	0.074
Equal-share	0.929	10.651	57.601	0.073
Shapley	0.911	10.125	57.445	0.073

equal-share, and Shapley benchmarks. Each method redistributes the same risk-savings pool using a weighting principle defined in Section 4.3, and performance is assessed through the metrics in (41)–(45). The results, averaged over all operating hours, are summarized in Table 2.

The tail-consistent allocation attains the highest TAC value, indicating an almost perfect correspondence with the risk structure embedded in the RU-CVaR objective. The all-scenarios rule follows closely, with minor deviations caused by including non-tail outcomes in the attribution. Capacity-based, equal-share, and energy-based schemes exhibit intermediate TAC levels, whereas the Shapley formulation shows the weakest alignment, since its coalition-value perspective does not directly reflect the tail-loss contributions driving the risk-averse objective.

Regarding IRmin, the capacity-based and all-scenarios rules yield slightly larger values due to their inherent averaging effects, which tend to smooth individual differences. By contrast, the tail-consistent rule does not seek to maximize protection for the least-advantaged participant; allocations are determined strictly by tail-risk contributions. As a result, the weakest unit is not explicitly subsidized, but the incentive structure remains behaviorally consistent and economically grounded, rewarding participants in proportion to their effectiveness in mitigating extreme risks.

In terms of equity, the tail-consistent rule achieves the lowest EFmax and the smallest Gini coefficient, indicating that payout disparities across members are effectively contained. The normalized NEF is identical across all schemes and is therefore not discussed further. Proportional and equal-share baselines deliver more uneven outcomes, as they ignore asymmetries in individual risk exposure, while the Shapley rule only partially attenuates this imbalance and remains less consistent with a tail-risk-based allocation principle.

Overall, these results show that the tail-consistent allocation offers the most balanced performance across all evaluation dimensions, confirming that aligning redistribution with tail-loss responsibility yields outcomes that are both efficient and equitable in risk-averse renewable aggregation. Fig. 10 further illustrates this behavior for a representative hour, where the tail-based scheme assigns larger risk-saving units to the most exposed wind farms while avoiding excessive compensation to low-risk members, thereby directing benefits proportionally to tail-risk contributions in practical operation. Since the proposed allocation relies on tail-responsibility weights computed from a finite scenario set, their numerical stability under different scenario sizes is further examined in the following subsection.

5.5. Stability of tail-responsibility weights under different scenario sizes

A supplementary scenario-size stability test is conducted to verify whether the tail-responsibility weights remain numerically stable under finite scenario sampling. Since the proposed settlement rule redistributes the risk-savings pool according to the tail-responsibility weights, this test is used to examine whether sampling noise in the tail scenarios may materially affect the resulting allocation. Proposition 3 concerns the convergence of the aggregate CVaR quantity, but it does not by itself guarantee the stability of the normalized tail-responsibility weights. Therefore, the present test is introduced as an empirical stability check for the allocation weights themselves.

The case with the largest scenario size ($S = 1200$) is used as the reference. The tail-responsibility vectors obtained under smaller scenario sets are compared only for groups whose memberships remain identical to the reference case, so that the effect of changing group composition is excluded from the comparison. In this way, the test isolates the numerical stability of the tail-responsibility weights conditional on fixed group composition, rather than the full end-to-end stability of the jointly optimized grouping-and-allocation mechanism.

For each matched group, two deviation measures are used. The first is the L_1 deviation of the normalized tail-responsibility vector,

$$\varepsilon_\omega^{(1)}(g,h;S) = \frac{1}{2}\sum_{i\in\mathcal{I}} \left|\omega_{i,g,h}^{(S)} - \omega_{i,\bar{g},h}^{(\mathrm{ref})}\right|,$$

where $\bar{g}$ denotes the matched reference group. This indicator measures the aggregate deviation of the full weight vector from the reference case. The second is the maximum single-member deviation,

$$\varepsilon_\omega^{(\infty)}(g,h;S) = \max_{i\in\mathcal{I}} \left|\omega_{i,g,h}^{(S)} - \omega_{i,\bar{g},h}^{(\mathrm{ref})}\right|.$$

This indicator measures the largest local deviation among all members. Since the first metric aggregates deviations over all members while the second reports only the largest componentwise deviation, $\varepsilon_\omega^{(1)}$ can naturally be larger than $\varepsilon_\omega^{(\infty)}$. The reported values are obtained by averaging these deviations over all matched groups and hours.

As shown in Fig. 11, both deviation indicators exhibit a clear overall downward trend as the scenario size increases. The main reduction occurs when the scenario set increases from a small sample to a moderate sample, which indicates that the sampling noise in tail identification and tail-responsibility estimation is substantially reduced once the model moves beyond the small-sample regime. This trend should not be

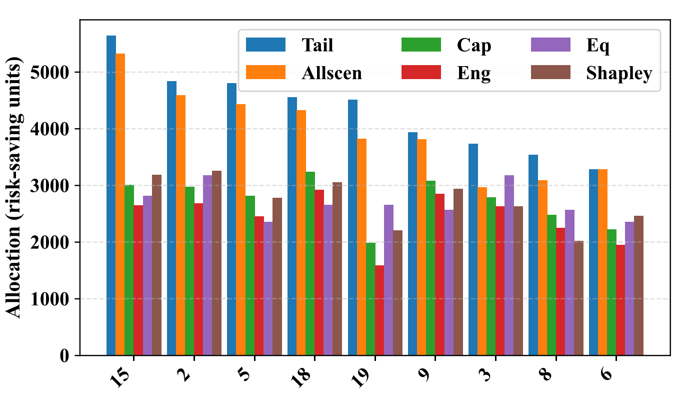

Fig. 10. Allocation comparison across methods at a representative hour ($h = 1$).

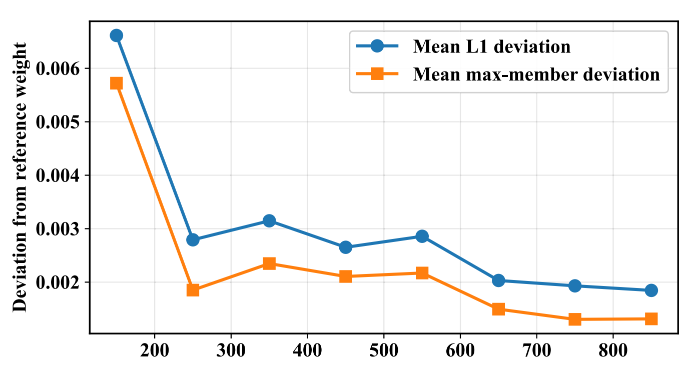

Fig. 11. Stability of tail-responsibility weights across different scenario sizes.

interpreted as implying a trade-off-type optimal scenario size; instead, any minor local irregularity is attributed to finite-sample fluctuations caused by independently re-optimized instances at each scenario size. For larger scenario sizes, both indicators stay at a low level, indicating that the tail-responsibility weights become practically stable once a moderate-to-large scenario set is used.

5.6. Parameter sensitivity analysis

A joint sensitivity analysis is conducted to examine how the risk-aversion weight λ and the CVaR confidence level α affect the advantage of grouping. In the CVaR-based objective, λ determines the trade-off between expected profit and downside tail risk. When λ increases, more emphasis is placed on reducing extreme losses, and a larger sacrifice of expected profit can be accepted to reduce tail exposure. When λ decreases, more emphasis is placed on expected profit, and a higher tolerance for tail risk is implied. In addition, α determines which tail outcomes are emphasized. For example, $\alpha = 0.95$ focuses on the worst 5% scenarios, while a larger α concentrates on rarer and more extreme outcomes. Therefore, λ and α are expected to jointly produce a trade-off. When they are too small, tail risk can be underweighted, while when they are too large, bidding decisions can become overly conservative.

To quantify robustness under different risk preferences, a normalized risk-adjusted performance gain is evaluated on a two-dimensional grid of (λ, α). For each parameter pair, both the grouped-aggregated model and the individual-bidding benchmark are solved under the same scenario set. Then the following metric is computed:

$$\Delta(\lambda, \alpha) = \frac{J^{\text{group}}(\lambda, \alpha) - J^{\text{ind}}(\lambda, \alpha)}{\left|J^{\text{ind}}(\lambda, \alpha)\right|}, \tag{49}$$

where $J(\lambda, \alpha)$ denotes the risk-adjusted utility that is consistent with the objective in (13). It combines the expected profit with the λ-weighted tail-risk term evaluated at confidence level α. When $\Delta(\lambda, \alpha)$ increases, grouping provides a stronger improvement over individual bidding under that risk preference.

Fig. 12 shows that $\Delta(\lambda, \alpha)$ stays positive over a broad region. This indicates that the grouped-aggregated scheme outperforms individual bidding under a wide range of risk preferences. The largest gains appear when λ takes moderate-to-high values and α stays around 0.94 to 0.95. Under these settings, tail-risk reduction receives sufficient emphasis while bids do not become overly conservative. When λ becomes very small, the tail-risk term receives limited weight, and the benefit from uncertainty compensation becomes less pronounced in the risk-adjusted metric. When α becomes very large, the objective places most emphasis on extremely rare outcomes, and bids can become more conservative, which reduces the incremental gain relative to the individual benchmark. Overall, the joint analysis shows that the advantage of grouping does not rely on a single parameter choice, and robust improvements are maintained under realistic risk-preference settings.

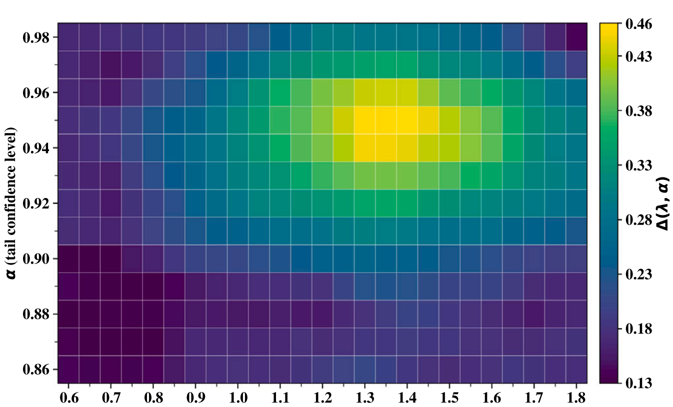

Fig. 12. Joint sensitivity heatmap of the normalized risk-adjusted utility improvement $\Delta(\lambda, \alpha)$.

6. Conclusion and future work

6.1. Conclusion

This paper develops a co-optimization framework that jointly determines wind-farm grouping and aggregated bidding for multiple wind farms under TOU exogenous day-ahead price and wind-speed uncertainty. By clustering wind farms according to their over- and under-generation risk characteristics under practical coordination constraints, the framework captures resource heterogeneity while ensuring feasible group-level operation. The objective function integrates asymmetric imbalance penalties with a CVaR-based tail-risk component, thus aligning day-ahead offers with the severity of imbalance exposure. Simulation results demonstrate that coordinated grouping and aggregation substantially enhance risk-adjusted performance compared with individual bidding. Sensitivity analyses further clarify the profit–risk mechanisms. The ex-post revenue allocation based on the proposed tail-consistent rule achieves a balanced trade-off between fairness and incentive compatibility, ensuring that portfolio-level efficiency gains are fairly transmitted to individual participants without unintended compensation among group members.

6.2. Future work

Several extensions are worth pursuing as follows: First, extending the current hour-separable day-ahead formulation to a multi-stage or rolling-horizon setting would allow forecast updates and balancing-stage recourse to be represented more faithfully, and would also enable multi-hour scenario trajectories to capture temporal persistence of low-wind events. Second, if shorter trading intervals (e.g., 15-min or 5-min) or stricter inter-hour deliverability are considered, linear inter-temporal constraints such as ramping limits can be incorporated while preserving a MILP structure. This is consistent with the fact that active-power controllability requirements for wind power plants are typically specified at sub-hour time scales in grid codes. Integrating additional flexibility like storage, hydrogen (P2G), and demand response/EVs may amplify diversification benefits and reshape settlement design. In particular, when energy-limited resources are introduced, inter-hour state variables and terminal energy-neutrality constraints can be added to prevent infeasible energy shifting and to provide more conservative estimates under sustained low-wind conditions. Furthermore, scalable and privacy-preserving algorithms, e.g., ADMM or federated variants, should be developed to enable deployment with limited data sharing across owners. Finally, field trials using utility/market data and probabilistic LMP forecasts would provide out-of-sample validation, quantify operational impacts (e.g., reserve usage and ramping), and calibrate penalty parameters for different markets. These directions aim to translate the demonstrated efficiency and fairness of group–aggregated bidding into robust, market-ready practice.

CRediT authorship contribution statement

Yiming Li: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Zhenwei Guo:** Writing – review & editing, Supervision, Conceptualization. **Zhen Wang:** Writing – review & editing. **Lin Wang:** Writing – review & editing. **Qinmin Yang:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Wenchao Meng:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary details for residual modeling and sampling

A.1. Johnson SU residual model and parameter interpretation

In the experiments, the wind-power forecast residual at wind farm i and hour h is modeled by a Johnson SU distribution. This family is adopted because it can represent non-Gaussian, skewed, and heavy-tailed residuals while still providing a closed-form quantile expression that enables stable inverse-transform sampling within the copula-based scenario generator.

The Johnson SU distribution is parameterized by γ, δ, ξ, and λ^{JSU}, where $\delta > 0$ and $\lambda^{\mathrm{JSU}} > 0$. Let

$$y = \frac{x-\xi}{\lambda^{\mathrm{JSU}}}.$$

Its cumulative distribution function is

$$F(x) = \Phi(\gamma + \delta\ \mathrm{asinh}(y)),$$

and the inverse-transform sampling formula is

$$F^{-1}(u) = \xi + \lambda^{\mathrm{JSU}} \sinh\left(\frac{\Phi^{-1}(u)-\gamma}{\delta}\right), \quad u \in (0,1).$$

For completeness, the probability density function is

$$f(x) = \frac{\delta}{\lambda^{\mathrm{JSU}}\sqrt{2\pi}\sqrt{1+y^2}} \exp\left[-\frac{1}{2}(\gamma + \delta\ \mathrm{asinh}(y))^2\right].$$

A convenient way to interpret the parameters is through the transformed-normal representation. Define

$$Z = \gamma + \delta\ \mathrm{asinh}\left(\frac{X-\xi}{\lambda^{\mathrm{JSU}}}\right).$$

Then $Z \sim \mathcal{N}(0,1)$. Conversely,

$$X = \xi + \lambda^{\mathrm{JSU}} \sinh\left(\frac{Z-\gamma}{\delta}\right).$$

Under this representation, ξ shifts the distribution along the horizontal axis and acts as a location parameter, while λ^{JSU} scales the overall spread and acts as a scale parameter. The parameters γ and δ jointly determine the shape of the distribution. They affect both asymmetry and tail thickness because they change how the underlying normal quantiles are mapped through the nonlinear $\sinh(\cdot)$ transformation.

The role of γ can be understood as a shift in the transformed-normal argument. When $\gamma \neq 0$, the mapping from the symmetric normal input becomes asymmetric after the nonlinear transformation, which leads to skewness in the resulting residual distribution. The role of δ can be understood as a stretching factor. For a fixed normal quantile z, the magnitude of $(z-\gamma)/\delta$ increases when δ decreases. Since $\sinh(t)$ grows approximately exponentially for large $|t|$, smaller δ generally produces heavier tails, while larger δ produces lighter tails. In practice, skewness and tail heaviness are jointly shaped by (γ, δ), and the pair $(\xi, \lambda^{\mathrm{JSU}})$ determines the location and the overall scale.

A.2. Parameter estimation and diagnostic check

For each wind farm i and operating hour h, an empirical residual series $\{r_{i,h,t}\}_{t=1}^{T}$ is obtained from the simulated true power and the constructed day-ahead forecast. The Johnson SU parameters $\theta_{i,h} = (\gamma_{i,h}, \delta_{i,h}, \xi_{i,h}, \lambda^{\mathrm{JSU}}_{i,h})$ are fitted by maximum likelihood estimation. Specifically, the log-likelihood is written as

$$\ell(\theta_{i,h}) = \sum_{t=1}^{T} \log f(r_{i,h,t}; \theta_{i,h}),$$

and the estimate is obtained by

$$\hat{\theta}_{i,h} \in \arg\max_{\theta_{i,h}}\ \ell(\theta_{i,h}) \quad \text{subject to} \quad \delta_{i,h} > 0,\ \lambda^{\mathrm{JSU}}_{i,h} > 0.$$

The constraints ensure a valid distribution and are enforced during numerical optimization.

A basic diagnostic check is conducted to ensure that the fitted Johnson SU captures both the central shape and the tails of the empirical residuals. In the revised manuscript, the fitting diagnostics and representative plots are reported in the Appendix, including a visual comparison between the empirical residual histogram and the fitted density, and a quantile-based check that focuses on tail quantiles that matter for the CVaR-based risk evaluation. These diagnostics are intended to confirm that the fitted marginals provide a reasonable heavy-tailed approximation for the residuals used in scenario generation.

A.3. Generation of correlated residual scenarios via Gaussian copula

This subsection summarizes how correlated residual scenarios are generated using a Gaussian copula together with the fitted Johnson SU marginals. The procedure is carried out for each operating hour h and produces a finite scenario set $\{r_{i,h,s}\}_{i\in\mathcal{I}, s\in\mathcal{S}}$ that preserves both the fitted marginal residual distribution of each wind farm and the prescribed cross-farm dependence structure. The required inputs for a fixed hour h include the cross-farm correlation matrix $\boldsymbol{R}$ defined in Section 2, the fitted marginal model $F_{\mathrm{JSU},i,h}$ for each wind farm i, and the point forecast $\hat{p}_{i,h}$ used in the day-ahead bidding evaluation.

Correlated Gaussian samples are first generated to impose the dependence implied by $\boldsymbol{R}$. In implementation, a square-root factor $\boldsymbol{L}$ is computed such that $\boldsymbol{L}\boldsymbol{L}^{\top} = \boldsymbol{R}$ (for instance via a Cholesky factorization when $\boldsymbol{R}$ is numerically positive definite). For each scenario $s \in \mathcal{S}$, an i.i.d. standard normal vector $\boldsymbol{\epsilon}_{h,s} \sim \mathcal{N}(\boldsymbol{0}, \mathbf{I})$ is drawn and mapped as $\mathbf{z}_{h,s} = \boldsymbol{L}\boldsymbol{\epsilon}_{h,s}$, which yields $\mathbf{z}_{h,s} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R})$. When $\boldsymbol{R}$ is near-singular due to finite-sample effects, a numerically stable factorization can be obtained by a standard eigenvalue correction before computing $\boldsymbol{L}$.

The dependence is then transferred from the Gaussian space to the fitted Johnson SU marginals through the probability integral transform. Each Gaussian component is mapped to a uniform variable by $u_{i,h,s} = \Phi(z_{i,h,s})$, and the residual realization is obtained by applying the fitted Johnson SU inverse CDF componentwise as $r_{i,h,s} = F^{-1}_{\mathrm{JSU},i,h}(u_{i,h,s})$. This construction preserves the fitted marginal residual behavior farm-wise while maintaining the Gaussian-copula dependence structure across farms. Finally, the corresponding power scenario is formed by adding residuals to the point forecast:

$$p_{i,h,s} = \hat{p}_{i,h} + r_{i,h,s}.$$

If needed for physical feasibility, the resulting values can be truncated to the admissible range $[0, C_i]$ in post-processing. This optional step is only used to enforce physical bounds and does not change the optimization formulation.

In the baseline setting used in the numerical experiments, all scenarios are assigned uniform weights, $w_s = 1/|\mathcal{S}|$, which is consistent with the finite-sample RU–CVaR formulation adopted in both the bidding optimization and the settlement-layer risk attribution. If non-uniform scenario weights are available (e.g., empirical frequencies or importance-sampling weights), the same formulation can incorporate them directly by replacing w_s accordingly (Table 3).

For convenience, the core sampling steps can be summarized as

Table 3
Main parameter settings used in the simulated data generation.

Parameter	Meaning	Setting / Range
N	Number of wind farms	20
ρ_0	Base spatial correlation	0.85
ℓ	Correlation length	60 km
(x_i, y_i)	Wind-farm location range	$[-100, 100]$ km
C_i	Installed-capacity range	80–150 MW
k_i	Weibull shape parameter of wind speed at wind farm i	2.2
λ_i^{base}	Baseline Weibull scale parameter of wind speed at wind farm i	8.0
λ_i^{amp}	Amplitude of diurnal modulation of the Weibull scale parameter at wind farm i	3.0
$h_{0,i}$	Phase-shift parameter of diurnal modulation at wind farm i	2
$\lvert S\rvert$	Scenario size in the baseline optimization case	250
Scenario weights	Probability assigned to each scenario	Uniform, $1/\lvert S\rvert$

$$\begin{aligned}\mathbf{z}_{h,s} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{R}) \rightarrow\ & u_{i,h,s} = \Phi(z_{i,h,s}) \\ \rightarrow\ & r_{i,h,s} = F^{-1}_{\text{JSU},i,h}(u_{i,h,s}) \\ \rightarrow\ & p_{i,h,s} = \hat{p}_{i,h} + r_{i,h,s},\ s \in \mathcal{S}.\end{aligned}$$

In the baseline setting, uniform scenario weights are used:

$$w_s = \frac{1}{|S|}.$$

Appendix B. Necessary proof of the tail-consistent allocation rule

B.1. Proof of Proposition 1

Proof. Conservation follows by summing (25) over i. For monotonicity, write $\delta U_{i,h,s} = \Delta U_{g,h,s}\, a_{i,h,s}/A_s$ with $A_s = \sum_j a_{j,h,s}$. Treating $\Delta U_{g,h,s}$ and $\{a_{j,h,s}\}_{j\neq i}$ as constants gives

$$\frac{\partial \delta U_{i,h,s}}{\partial a_{i,h,s}} = \Delta U_{g,h,s}\frac{A_s - a_{i,h,s}}{A_s^2} > 0$$

whenever $A_s > a_{i,h,s}$. The surplus case is identical. □

B.2. Proof of Proposition 2

Proof. For each scenario s, write $d_i(s) = q_{i,h} - p_{i,h,s}$. By convexity of $x \mapsto x_+$ and the triangle inequality, $\left(\sum_i d_i(s)\right)_+ \le \sum_i d_i(s)_+$ and $\left(-\sum_i d_i(s)\right)_+ \le \sum_i \left(-d_i(s)\right)_+$. Hence, with $Q^{\text{bench}}_{g,h} = \sum_i q_{i,h}$,

$$\Delta U_{g,h,s}\left(Q^{\text{bench}}_{g,h}\right) \le \sum_i (q_{i,h} - p_{i,h,s})_+,$$

$$\Delta O_{g,h,s}\left(Q^{\text{bench}}_{g,h}\right) \le \sum_i (p_{i,h,s} - q_{i,h})_+,$$

which implies the loss inequality

$$L_{g,h,s}\left(Q^{\text{bench}}_{g,h}\right) \le \sum_{i\in g} L^{\text{ind}}_{i,h,s}(q_{i,h}),\ \forall s.$$

By monotonicity and subadditivity of CVaR_α,

$$\begin{aligned}&\text{CVaR}_\alpha\left(\left\{L_{g,h,s}(Q^{\text{bench}}_{g,h})\right\}_s\right)\\ &\le \text{CVaR}_\alpha\left(\left\{\sum_i L^{\text{ind}}_{i,h,s}(q_{i,h})\right\}_s\right)\\ &\le \sum_i \text{CVaR}_\alpha\left(\left\{L^{\text{ind}}_{i,h,s}(q_{i,h})\right\}_s\right).\end{aligned}$$

By the CVaR-dominance assumption,

$$\text{CVaR}_\alpha\left(\{L_{g,h,s}(Q_{g,h})\}_s\right) \le \text{CVaR}_\alpha\left(\left\{L_{g,h,s}(Q^{\text{bench}}_{g,h})\right\}_s\right).$$

Combining the previous display with the dominance inequality yields

$$\text{CVaR}_\alpha\left(\{L_{g,h,s}(Q_{g,h})\}_s\right) \le \sum_i \text{CVaR}_\alpha\left(\left\{L^{\text{ind}}_{i,h,s}(q_{i,h})\right\}_s\right),$$

which is exactly $RS_{g,h} \ge 0$ by the definition in (28). □

B.3. Proof of Proposition 3

By definition,

$$x_{i,g,h} = \sum_{s\in T_{g,h}} L^{\text{contrib}}_{i,h,s},\ S_{g,h} = \sum_{j\in g} x_{j,g,h},$$

and

$$\omega_{i,g,h} = \frac{x_{i,g,h}}{S_{g,h}}.$$

Hence, the risk-savings allocation to member i can be written as

$$r^{\text{risk}}_{i,h} = \frac{x_{i,g,h}}{S_{g,h}} RS_{g,h}.$$

Fix all other members' tail-loss contributions and define

$$C_{g,h} = \sum_{j\in g,\, j\neq i} x_{j,g,h},$$

so that

$$S_{g,h} = x_{i,g,h} + C_{g,h}.$$

Then

$$r^{\text{risk}}_{i,h} = \frac{x_{i,g,h}}{x_{i,g,h} + C_{g,h}} RS_{g,h}(x_{i,g,h}).$$

Differentiating with respect to $x_{i,g,h}$ gives

$$\frac{\partial r^{\text{risk}}_{i,h}}{\partial x_{i,g,h}} = \frac{C_{g,h}}{(x_{i,g,h} + C_{g,h})^2} RS_{g,h} + \frac{x_{i,g,h}}{x_{i,g,h} + C_{g,h}} \frac{\partial RS_{g,h}}{\partial x_{i,g,h}}.$$

Since $C_{g,h} = S_{g,h} - x_{i,g,h}$, this can be rewritten as

$$\frac{\partial r^{\text{risk}}_{i,h}}{\partial x_{i,g,h}} = \frac{S_{g,h} - x_{i,g,h}}{S^2_{g,h}} RS_{g,h} + \frac{x_{i,g,h}}{S_{g,h}} \frac{\partial RS_{g,h}}{\partial x_{i,g,h}}.$$

Therefore, a sufficient condition for

$$\frac{\partial r^{\text{risk}}_{i,h}}{\partial x_{i,g,h}} \le 0$$

is

$$\frac{\partial RS_{g,h}}{\partial x_{i,g,h}} \le -\frac{S_{g,h} - x_{i,g,h}}{x_{i,g,h} S_{g,h}} RS_{g,h}.$$

Under this condition, $r^{\text{risk}}_{i,h}$ is non-increasing in $x_{i,g,h}$. Hence, increasing the member's own attributed tail-loss contribution cannot increase its own risk-savings payment. This proves the proposition. □

B.4. Proof of Proposition 4

Proof. Write

$$\phi(\tau) = \tau + \frac{1}{1-\alpha}\,\mathbb{E}\big[(L-\tau)_+\big],$$

$$\widehat{\phi}_S(\tau) = \tau + \frac{1}{1-\alpha}\cdot\frac{1}{S}\sum_{s=1}^{S}(L_s-\tau)_+,$$

and denote $\tau^\star = \arg\min_\tau \phi(\tau) = v_\alpha$, $\hat{\tau}_S = \arg\min_\tau \widehat{\phi}_S(\tau)$. Then $\mathrm{CVaR}_\alpha = \phi(\tau^\star)$ and $\widehat{\mathrm{CVaR}}_{\alpha,S} = \widehat{\phi}_S(\hat{\tau}_S)$.

(i) Uniform concentration (bounded differences). Fix τ. Replacing a single sample L_t by L'_t changes $\widehat{\phi}_S(\tau)$ by at most $\frac{1}{1-\alpha}\cdot\frac{1}{S}\left|(L_t-\tau)_+ - (L'_t-\tau)_+\right| \le \frac{B}{(1-\alpha)S}$, since $(\cdot-\tau)_+$ is 1-Lipschitz and $0 \le L \le B$. By McDiarmid's inequality,

$$\mathbb{P}\left(\left|\widehat{\phi}_S(\tau)-\phi(\tau)\right| \ge t\right) \le 2\exp\left(-\frac{2(1-\alpha)^2 S}{B^2}\,t^2\right).$$

A Lipschitz-in-τ covering argument (mesh $h \asymp S^{-1/2}$) yields, for any $\delta \in (0,1)$,

$$\mathbb{P}\left(\sup_{\tau\in\mathbb{R}}\left|\widehat{\phi}_S(\tau)-\phi(\tau)\right| \le \Delta_S\right) \ge 1-\delta,$$

$$\Delta_S = \frac{B}{1-\alpha}\sqrt{\frac{2\log(2/\delta)}{S}}.$$

(ii) Local strong convexity and argmin stability. Under the uniqueness and density assumptions, $\phi'(\tau) = 1 - \frac{1}{1-\alpha}\mathbb{P}(L\ge\tau)$ and $\phi''(\tau) = \frac{1}{1-\alpha}f_L(\tau)$ imply that there exist $r, m > 0$ such that $|\tau - v_\alpha| \le r \to \phi''(\tau) \ge \kappa = \frac{m}{1-\alpha} > 0$. Thus ϕ is κ-strongly convex near $\tau^\star$. Combining optimality $\widehat{\phi}_S(\hat{\tau}_S) \le \widehat{\phi}_S(\tau^\star)$ with the uniform bound in (i) gives $\phi(\hat{\tau}_S)-\phi(\tau^\star) \le 2\Delta_S$. Strong convexity then yields $|\hat{\tau}_S-\tau^\star| \le \sqrt{4\Delta_S/\kappa} = O\left((1-\alpha)^{-1/2}S^{-1/4}\right)$. Refining via the first-order conditions $\widehat{\phi}'_S(\hat{\tau}_S) = \phi'(\tau^\star) = 0$ and a mean-value expansion of ϕ' shows in fact $|\hat{\tau}_S - \tau^\star| = O_{\mathbb{P}}\left((1-\alpha)^{-1}S^{-1/2}\right)$.

(iii) Deviation of $\widehat{\mathrm{CVaR}}_{\alpha,S}$. Decompose

$$\widehat{\phi}_S(\hat{\tau}_S) - \phi(\tau^\star) = \underbrace{\left[\widehat{\phi}_S(\hat{\tau}_S) - \phi(\hat{\tau}_S)\right]}_{\le \Delta_S} + \underbrace{\left[\phi(\hat{\tau}_S) - \phi(\tau^\star)\right]}_{\le \frac{1}{2}\sup\phi''\cdot|\hat{\tau}_S-\tau^\star|^2}.$$

The first term is bounded by Δ_S from (i). The second is of order $O_{\mathbb{P}}\left((1-\alpha)^{-1}S^{-1}\right)$ because $\sup\phi'' \le (1-\alpha)^{-1}\sup f_L$ on a neighborhood of v_α and $|\hat{\tau}_S-\tau^\star| = O_{\mathbb{P}}((1-\alpha)^{-1}S^{-1/2})$. Absorbing constants yields (46).

(iv) Central limit theorem. Define

$$\psi(L) = \left((L-v_\alpha)_+ - \mathbb{E}[(L-v_\alpha)_+]\right)/(1-\alpha).$$

A von Mises expansion of the functional $T(F) = \min_\tau\{\tau + \frac{1}{1-\alpha}\int(x-\tau)_+\,dF(x)\}$ around F gives

$$\widehat{\mathrm{CVaR}}_{\alpha,S} - \mathrm{CVaR}_\alpha = \frac{1}{S}\sum_{s=1}^{S}\psi(L_s) + o_{\mathbb{P}}(S^{-1/2}),$$

since the envelope condition $\phi'(\tau^\star) = 0$ removes the first-order effect of $\hat{\tau}_S$. By the classical CLT,

$$\sqrt{S}\left(\widehat{\mathrm{CVaR}}_{\alpha,S} - \mathrm{CVaR}_\alpha\right) \to \mathcal{N}\left(0, \mathrm{Var}(\psi(L))\right),$$

which equals (47). □

Data availability

Data will be made available on request.

References

[1] Afshar K, Ghiasvand FS, Bigdeli N. Optimal bidding strategy of wind power producers in pay-as-bid power markets. Renew Energy 2018;127:575–86. https://doi.org/10.1016/j.renene.2018.05.015

[2] Aghamohamadi M, Mahmoudi A, Haque MH. Adaptive robust recourse-based bidding strategy and capacity allocation of PV-WT-BES owning prosumers under uncertainties. IEEE Trans Ind Appl 2021;57:4170–86. https://doi.org/10.1109/TIA.2021.3072603

[3] Aghdam FH, Mohammadi-Ivatloo B, Abapour M, Marzband M, Rasti M, Pongracz E. Enhancing the risk-oriented participation of wind power plants in day-ahead, balancing, and hydrogen markets with shared multi-energy storage systems. J Energy Storage 2025;107:114911. https://doi.org/10.1016/j.est.2024.114911

[4] Ahunbay M Ş, Bichler M, Dobos T, Knörr J. Solving large-scale electricity market pricing problems in polynomial time. Eur J Oper Res 2024;318:605–17. https://doi.org/10.1016/j.ejor.2024.05.020

[5] Baringo L, Conejo AJ. Offering strategy of wind-power producer: a multi-stage risk-constrained approach. IEEE Trans Power Syst 2016;31:1420–9. https://doi.org/10.1109/TPWRS.2015.2411332

[6] Carrión M, Domínguez R, Oggioni G. Optimal participation of wind power producers in a hybrid intraday market: a multi-stage stochastic approach. Energy Econ 2025;144:108303. https://doi.org/10.1016/j.eneco.2025.108303

[7] Chen Q, Lyu R, Guo H, Su X. Real-time operation strategy of virtual power plants with optimal power disaggregation among heterogeneous resources. Appl Energy 2024;361:122876. https://doi.org/10.1016/j.apenergy.2024.122876

[8] Dai T, Qiao W. Optimal bidding strategy of a strategic wind power producer in the short-term market. IEEE Trans Sustain Energy 2015;6:707–19. https://doi.org/10.1109/TSTE.2015.2406322

[9] Dimitriadis CN, Tsimopoulos EG, Georgiadis MC. Co-optimized trading strategy of a renewable energy aggregator in electricity and green certificates markets. Renew Energy 2024;236:121444. https://doi.org/10.1016/j.renene.2024.121444

[10] Dirin S, Rahimiyan M, Baringo L. Optimal offering strategy for wind-storage systems under correlated wind production. Appl Energy 2023;333:120552. https://doi.org/10.1016/j.apenergy.2022.120552

[11] Farhadi A, Hajiaghayi M, Latifian M, Seddighin M, Yami H. Almost envy-freeness, envy-rank, and nash social welfare matchings. Proc AAAI Conf Artif Intell 2021;35:5355–62. https://doi.org/10.1609/aaai.v35i6.16675

[12] Guo Z, Pinson P, Chen S, Yang Q, Yang Z. Chance-constrained peer-to-peer joint energy and reserve market considering renewable generation uncertainty. IEEE Trans Smart Grid 2021;12:798–809. https://doi.org/10.1109/TSG.2020.3019603

[13] Ju L, Bai X, Li G, Gan W, Qi X, Ye F. Two-stage robust transaction optimization model and benefit allocation strategy for new energy power stations with shared energy storage considering green certificate and virtual energy storage mode. Appl Energy 2024;362:122996. https://doi.org/10.1016/j.apenergy.2024.122996

[14] Jung C, Schindler D. Wind speed distribution selection – a review of recent development and progress. Renew Sustain Energy Rev 2019;114:109290. https://doi.org/10.1016/j.rser.2019.109290

[15] Li H, Li X, Zhang Y, Zhao Y, Pan J, Zhao H. Declaration strategy of wind power and pumped storage participating in the power market considering multiple uncertainties. Energy 2024;293:130624. https://doi.org/10.1016/j.energy.2024.130624

[16] Li W, Dong F, Ji Z, Wang P. Internal and external coordinated distributionally robust bidding strategy of virtual power plant operator participating in day-ahead electricity spot and peaking ancillary services markets. Appl Energy 2025;386:125514. https://doi.org/10.1016/j.apenergy.2025.125514

[17] Lyu R, Guo H, Zheng K, Sun M, Chen Q. Co-optimizing bidding and power allocation of an EV aggregator providing real-time frequency regulation service. IEEE Trans Smart Grid 2023;14:4594–606. https://doi.org/10.1109/TSG.2023.3252664

[18] Meng F, Zhao D, Zhang X. A fair consensus adjustment mechanism for large-scale group decision making in term of gini coefficient. Eng Appl Artif Intell 2023;126:106962. https://doi.org/10.1016/j.engappai.2023.106962

[19] Moradi M, Farzaneh H. Demand response programs in decentralized hybrid local energy markets: evaluating the impact of risk-adjusted behavior of market players and the integration of renewable energy sources, using a novel bi-level optimization framework. Appl Energy 2025;390:125806. https://doi.org/10.1016/j.apenergy.2025.125806

[20] Qu K, Chen Y, Xie S, Zheng X, Zhu J. Segmented distributionally robust optimization for real-time power dispatch with wind uncertainty. IEEE Trans Power Syst 2024;39:2970–83. https://doi.org/10.1109/TPWRS.2023.3303313

[21] Rockafellar RT, Uryasev S, et al. Optimization of conditional value-at-risk. J Risk 2000;2:21–42.

[22] Rostami Ghadi F, Wong K-K, Javier López-Martínez F, Chae C-B, Tong K-F, Zhang Y. A Gaussian copula approach to the performance analysis of fluid antenna systems. IEEE Trans Wirel Commun 2024;23:17573–85. https://doi.org/10.1109/TWC.2024.3454558

[23] Shafiekhani M, Ahmadi A, Homaee O, Shafie-Khah M, Catalão JPS. Optimal bidding strategy of a renewable-based virtual power plant including wind and solar units and dispatchable loads. Energy 2022;239:122379. https://doi.org/10.1016/j.energy.2021.122379

[24] Sharma KC, Bhakar R, Tiwari HP. Strategic bidding for wind power producers in electricity markets. Energy Convers Manag 2014;86:259–67. https://doi.org/10.1016/j.enconman.2014.05.002

[25] Sun S, Liu Y, Tang Z, Tu M, Du X, Liu J. A hybrid stochastic-robust bidding model for wind-storage system in day-ahead market considering risk preference. Electr Power Syst Res 2025;238:111148. https://doi.org/10.1016/j.epsr.2024.111148

[26] Wang L, Liu R, Zeng W, Zhang L, Peng H, Calautit JK, Ma B, Zhang R, Ma X, Li X. Revealing the theoretical wind potential of the Qinghai-Tibet plateau: a novel Bayesian Monte-Carlo framework for the Weibull bivariate distribution. Energy Convers Manag 2025;325:119375. https://doi.org/10.1016/j.enconman.2024.119375

[27] Wang S, Xie S, Meng W, Yang Q. Power regulation and load mitigation of fixed-bottom offshore wind turbines via adaptive dynamic programming. IEEE Trans Syst Man Cybern Syst 2025;55:8018–29. https://doi.org/10.1109/TSMC.2025.3602152

[28] Wang Y, Wu X, Liu M, Zhang C, Wang H, Yue Y, Luo X. Bidding strategy of the virtual power plant considering green certificates and carbon trading. Energy Rep 2023;9:73–84. https://doi.org/10.1016/j.egyr.2023.04.042

[29] Wozabal D, Rameseder G. Optimal bidding of a virtual power plant on the spanish day-ahead and intraday market for electricity. Eur J Oper Res 2020;280:639–55. https://doi.org/10.1016/j.ejor.2019.07.022

[30] Yan Y, Sun M, Guo Z. How do carbon cap-and-trade mechanisms and renewable portfolio standards affect renewable energy investment? Energy Policy 2022;165:112938. https://doi.org/10.1016/j.enpol.2022.112938

[31] Yang X, Fan L, Li X, Meng L. Day-ahead and real-time market bidding and scheduling strategy for wind power participation based on shared energy storage. Electr Power Syst Res 2023;214:108903. https://doi.org/10.1016/j.epsr.2022.108903

[32] Yu Y, Guan Y, Tarek B, Leon MAL, Özcan KO, Feleke S, Anteneh D, Khan B, Vasquez JC, Guerrero JM. A review of international grid codes for wind power integration. Energy Convers Manag X 2025;28:101278. https://doi.org/10.1016/j.ecmx.2025.101278

[33] Zhang Q, Wu X, Deng X, Huang Y, Li C, Wu J. Bidding strategy for wind power and large-scale electric vehicles participating in day-ahead energy and frequency regulation market. Appl Energy 2023;341:121063. https://doi.org/10.1016/j.apenergy.2023.121063

[34] Zhang R, Jiang T, Li F, Li G, Chen H, Li X. Coordinated bidding strategy of wind farms and power-to-gas facilities using a cooperative game approach. IEEE Trans Sustain Energy 2020;11:2545–55. https://doi.org/10.1109/TSTE.2020.2965521

[35] Zheng QP, Wang J, Liu AL. Stochastic optimization for unit commitment—a review. IEEE Trans Power Syst 2015;30:1913–24. https://doi.org/10.1109/TPWRS.2014.2355204